ECE330 Quiz3

完整题目、你的完整回答与对应解答

共3题，逐题保留实际网页记录与上传手写原图。

每题按原页连续截图、完整手写图、解答的顺序排列。

长网页分段连续展示，相邻页留有重叠，防止截断文字。

手写图使用完整上传图片，未裁去纸面上的作答。

历史上传图片不同的版本均保留；相同图片不重复放大。

截图与数值记录来自你的已完成测验页面。

解答为独立解析，系统答案与反馈在原页截图中保留。

截图日期 2026年11月4日

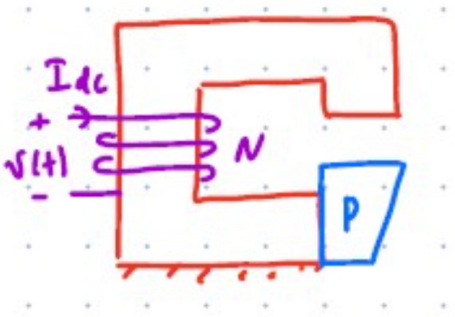
Quiz3 第1题 原始页面 1/3

完整题目、填写值、上传回答与系统记录；相邻页有重叠

Question 1

NOTE: Full credit (100%) for this question is 8 points: 5 points for the correct numerical answer(s) and 3 points for the correct conceptual answer. Points for the conceptual answer will be added manually.

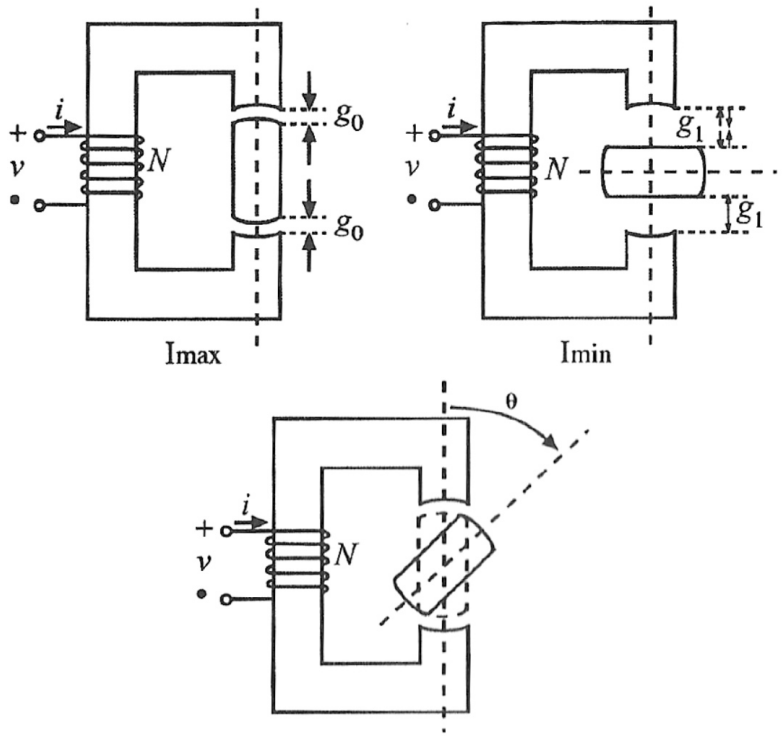
(3 pts) Consider the structure shown below.



The structure has infinite permiability, a cross sectional area A, a gap size x(t), a movable member P (restricted to moving only upward or downward), and the coil with N turns carrying a positive constant current I_{dc}. We have calculated the λ - i relationship as following: $\lambda = \frac{\mu_0 A N^2 I_{dc}}{x}$

Suppose at t=0 seconds, the member P is moving upward at a rate of 1 m/s. At time $t = 0$ seconds, is the instantaneous voltage across the coil positive, negative, or zero (with respect to the voltage polarity marked in the diagram)? Justify your choice. **Note: Points are awarded only for the justification. No points are awarded for the choice itself. Submit your answer on scratch paper.**

(5 pts) Consider the reluctance motor shown below.



The inductance for the reluctance motor with $N = 247$ turns can be approximated as a sinusoidal function

$L = L_1 + L_2 \cos(2\theta)$

Quiz3 第1题 原始页面 2/3

完整题目、填写值、上传回答与系统记录；相邻页有重叠

$$L = L_1 + L_2 \cos(2\theta)$$

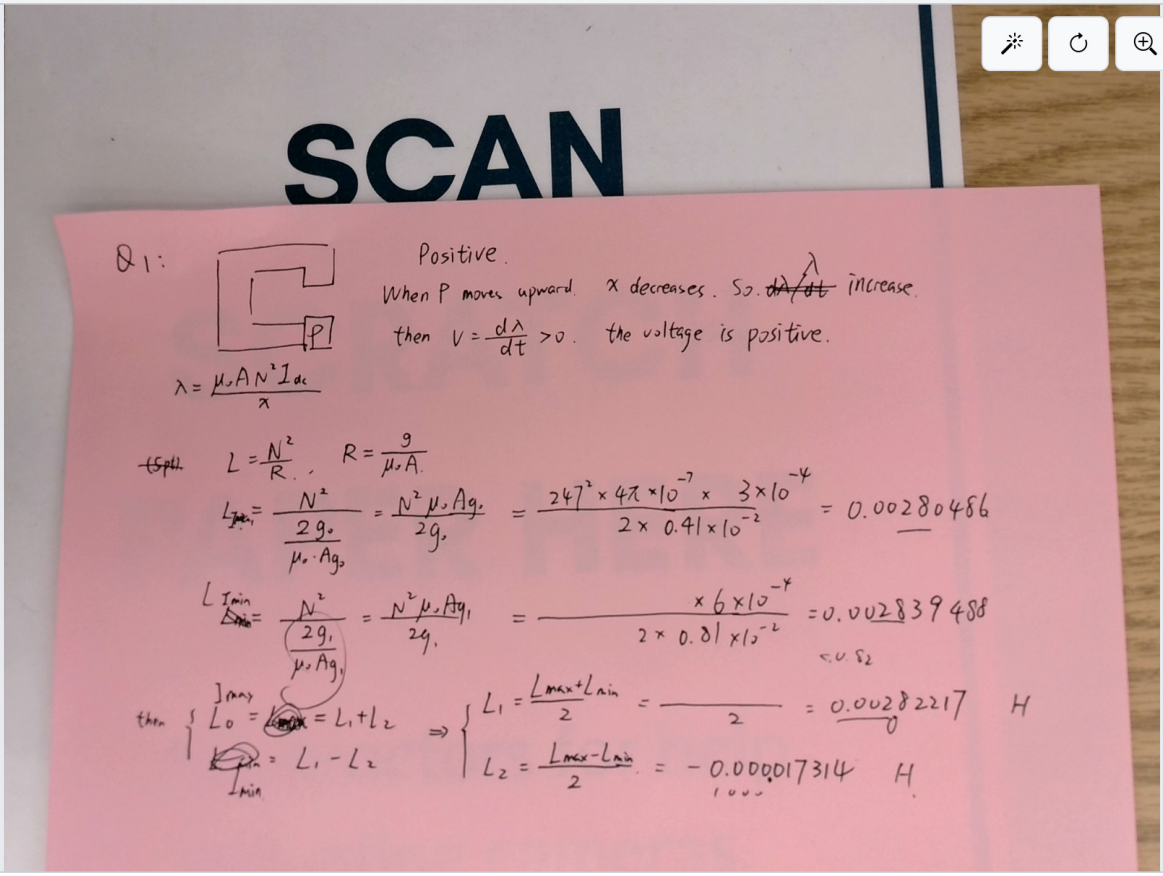
where the rotor angle θ is oriented as in the figure.

When $g_0 = 0.41$ cm, the cross-sectional area $A_{g0} = 3$ cm^2. When $g_1 = 0.81$ cm, the cross-sectional area $A_{g1} = 6$ cm^2.

What are the values of L_1 and L_2 for this motor? The reluctance of the iron core can be neglected.

L_1[mH]=	2.282217	?	✕ 0%
L_2[mH]=	-0.017314	?	✓ 100%

Image capture



Correct answer

L_1 [mH]= 2.8222 L_2 [mH]= −0.0173

Feedback from the Course Staff

Hide

Numerical: Partial credit from appeal

Submitted answer

yitaoma2@illinois.edu submitted at 2026-10-20 20:18:31 (CDT)

manual grading: 110%

auto-grading: 50%

Hide

Quiz3 第1题 原始页面 3/3

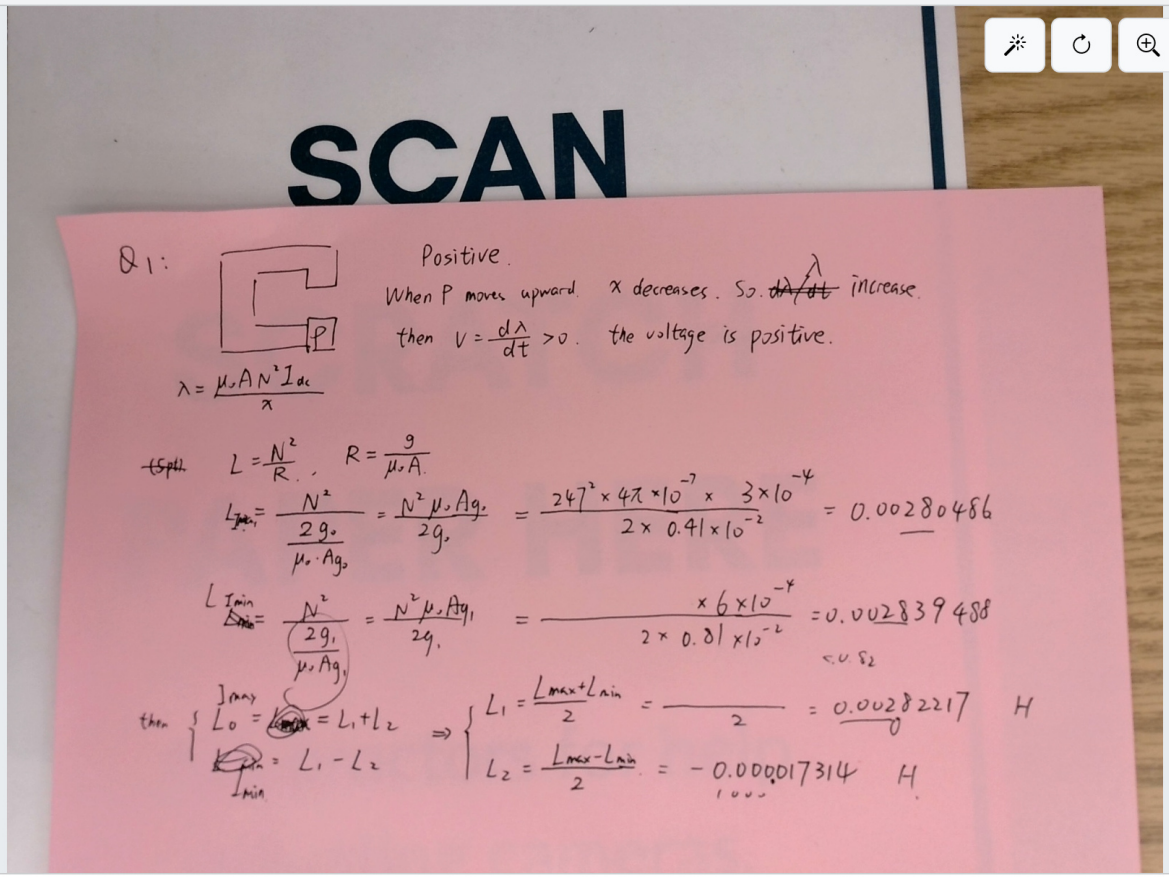
完整题目、填写值、上传回答与系统记录；相邻页有重叠

yitaoma2@illinois.edu submitted at 2026-10-20 20:18:31 (CDT) auto-grading: 50%

Hide

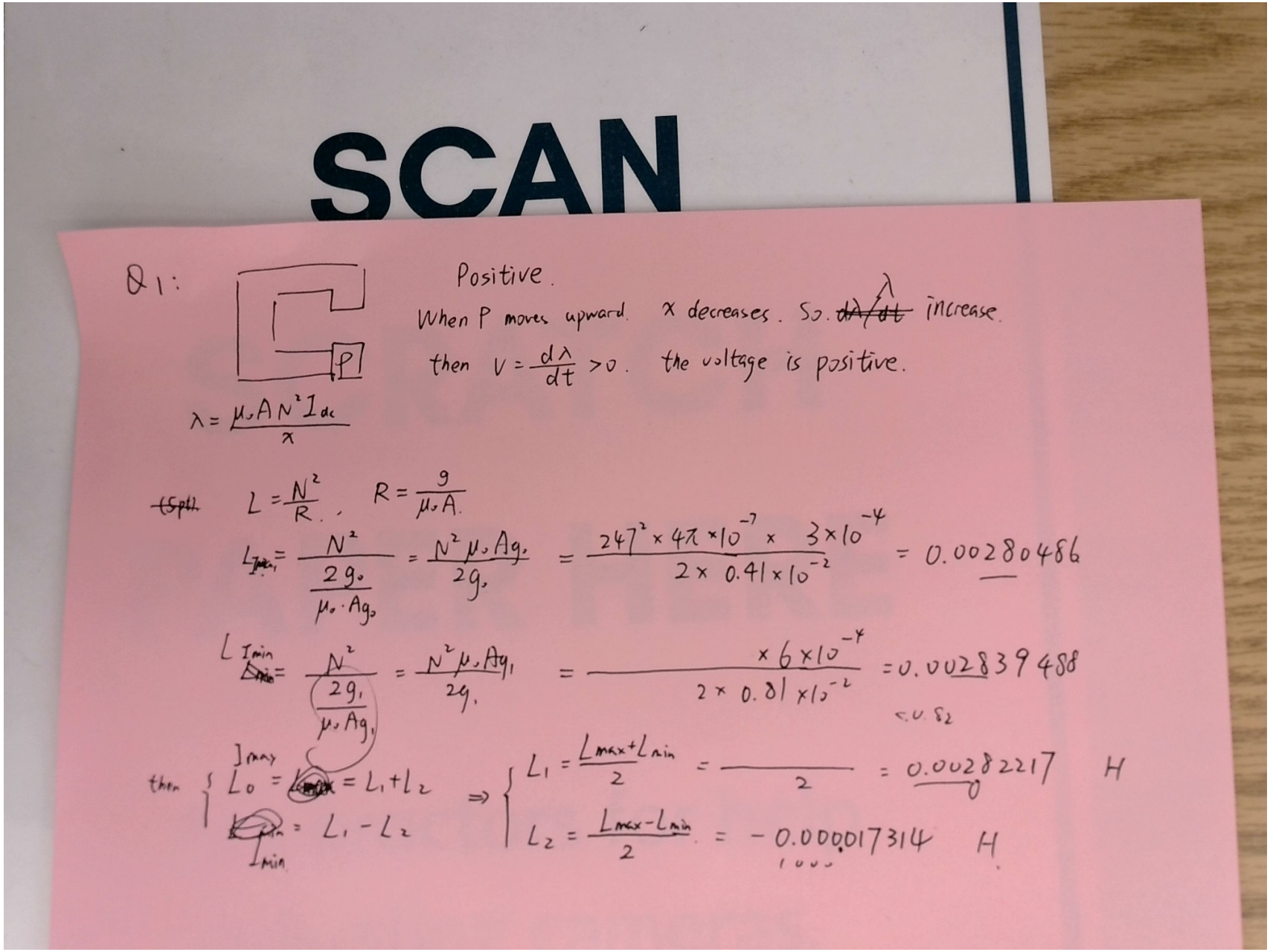
L_1 [mH]= 2.282217 ✕ 0% L_2 [mH]= −0.017314 The correct answer used for grading was -0.0173. ✓ 100%

Image capture



Quiz3 第1题 你的完整手写回答 1/1

当前最终回答原图



Quiz3 Q1 | Full uploaded image | 2000 x 1500 px

第1题 运动电压与磁阻电机

恒流时的运动电压符号、双气隙磁阻、角度电感的两个系数

第一问（3分） 铁芯磁导率无限大，截面积 A，线圈 N 匝，通正的恒定电流 I_dc。给定磁链关系如下；衔铁 P 只可上下移动，在 t＝0 时以 1 m/s 上移。按图中电压极性判断电压正、负或零，并说明理由。题目明确只为理由给分。

$$\lambda = \frac{\mu_0 A N^2 I_{dc}}{x}, \qquad v = \frac{d\lambda}{dt} = -\frac{\mu_0 A N^2 I_{dc}}{x^2}\dot{x}.$$

上移使气隙减小，所以 **位置速度为负，前面的负号与它相乘得到正电压**。恒流只使电流变化项为零。

第二问（5分） 磁阻电机 N＝247 匝；忽略铁芯磁阻。给定 $L(\theta)=L_1+L_2\cos(2\theta)$。图示 θ＝0 时每个气隙 $g_0=0.41$ cm，面积 A_g0＝3 cm²；另一极位每个气隙 $g_1=0.81$ cm，面积 A_g1＝6 cm²。求 L_1、L_2。每个极位的两气隙串联。

$$\begin{aligned}
L(0) &= \frac{\mu_0 N^2 A_{g0}}{2g_0} = \frac{(4\pi\times10^{-7})(247)^2(3\times10^{-4})}{2(0.0041)} \approx 2.80486\ \text{mH},\\
L\left(\frac{\pi}{2}\right) &= \frac{\mu_0 N^2 A_{g1}}{2g_1} = \frac{(4\pi\times10^{-7})(247)^2(6\times10^{-4})}{2(0.0081)} \approx 2.83949\ \text{mH},\\
L_1 &= \frac{L(0)+L(\frac{\pi}{2})}{2} \approx 2.82217\ \text{mH},\\
L_2 &= \frac{L(0)-L(\frac{\pi}{2})}{2} \approx -0.0173134\ \text{mH}.
\end{aligned}$$

符号检查 此题两极位截面积也变了；按实际参数，θ＝0 的电感略小，所以 L_2 为负。原图的 Imax/Imin 标签不能代替代入后的比较。

系统显示 **$L_1=2.8222$ mH、$L_2=-0.0173$ mH**。你的上传手写已有正确 L_1，输入栏却为 2.282217 mH；L_2 已判对。

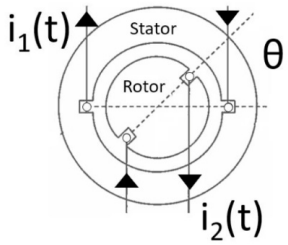
Quiz3 第2题 原始页面 1/4

完整题目、填写值、上传回答与系统记录；相邻页有重叠

Question 2

NOTE: Full credit (100%) for this question is 8 points: 5 points for the correct numerical answer(s) and 3 points for the correct conceptual answer. Points for the conceptual answer will be added manually.

A diesel generator is modeled as an electromechanical system shown below with two electrical and one mechanical ports. The stator is modeled as "terminal 1", the rotor is modeled as "terminal 2", and the angle between the stator and the rotor is modeled as the angle variable θ.



The rotor is connected to a DC current source of fixed 8 A, and driven by a diesel engine at a fixed angular velocity of 358 rad/s. The stator is left in open circuit. The $\lambda - i$ relations have been modeled as the following:

$$\begin{bmatrix}\lambda_1\\ \lambda_2\end{bmatrix} = \begin{bmatrix}9+1\cos(2\theta) & 0.22\cos(\theta)\\ 0.22\cos(\theta) & 66+34\cos(2\theta)\end{bmatrix}\begin{bmatrix}i_1\\ i_2\end{bmatrix}$$

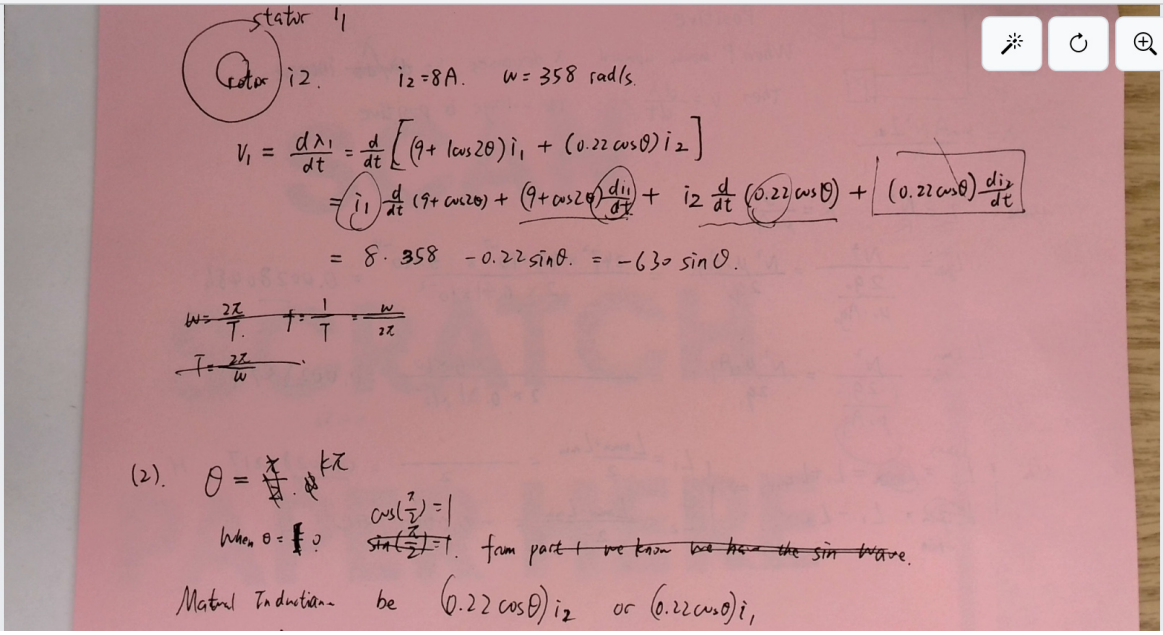
We observe a *sinusoidal* voltage waveform at the open-circuited stator.

(5 pts) What is the magnitude and frequency of the associated AC voltage phasor? (Reminder: your answer should be stated in RMS)

$\lvert\overline{V}_1\rvert$[V] =	630	?	✕ 0%
f[Hz] =	1.6	?	✕ 0%

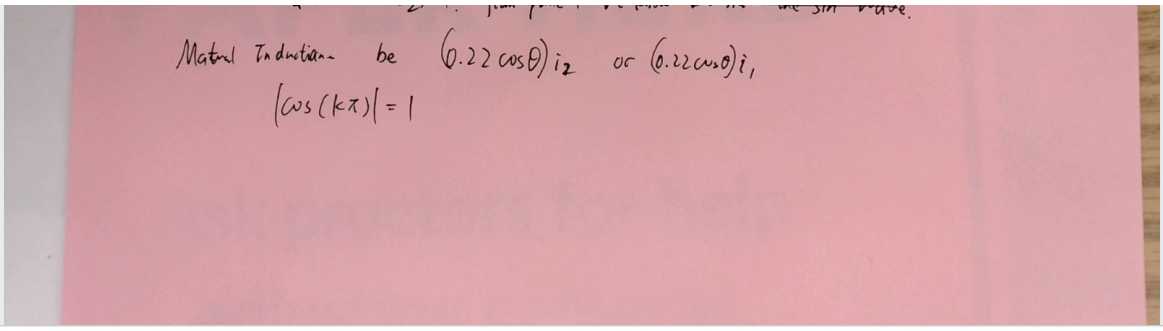
(3 pts) At which angle θ is the mutual inductance maximum? Explain your answer. **Note: Points are awarded only for the explanation. No points are awarded for the answer itself. Submit your answer on scratch paper.**

Image capture



Quiz3 第2题 原始页面 2/4

完整题目、填写值、上传回答与系统记录；相邻页有重叠



Correct answer

$|\overline{V}_1|$ [V] = 446 f [Hz] = 57.0

Feedback from the Course Staff (for submitted answer 3)

Hide

Numerical: See feedback in gradescope Conceptual: Partially incorrect solution (-1.5 pt)

Submitted answer 3

yitaoma2@illinois.edu submitted at 2026-10-20 20:47:25 (CDT)

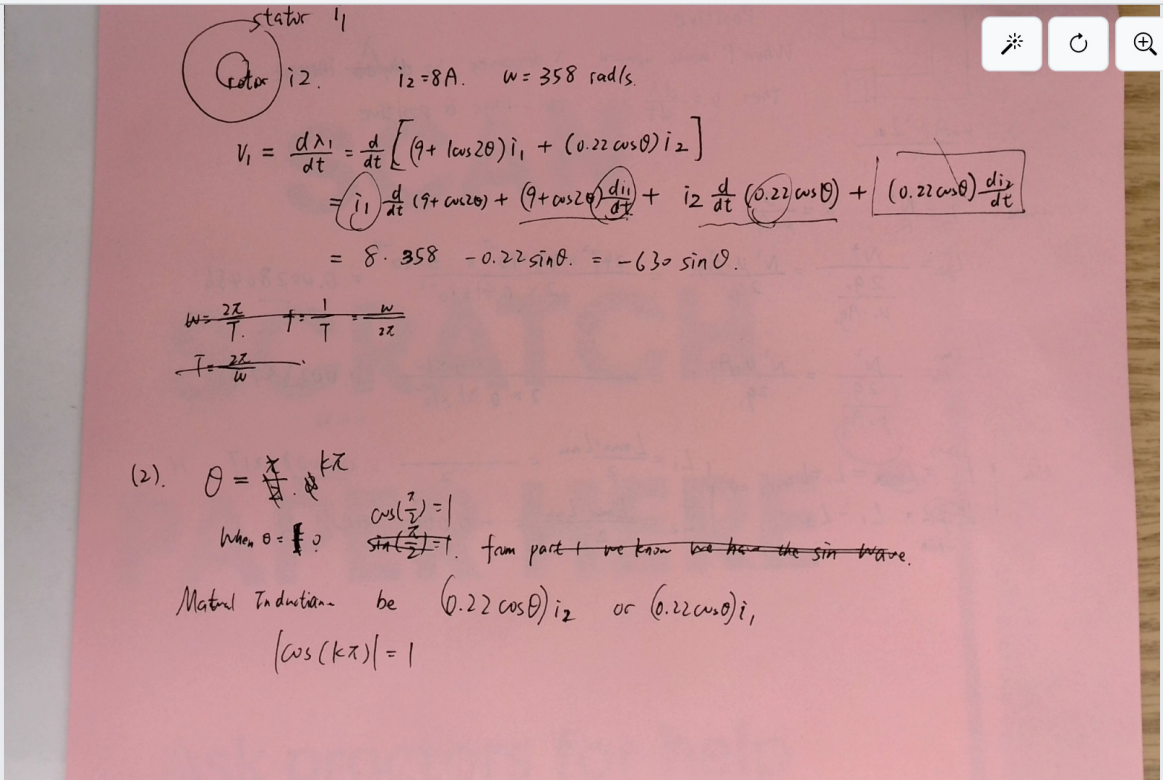
manual grading: 90%

auto-grading: 0%

Hide

$|\overline{V}_1|$ [V] = 630 | Your answer does not have precision within the specified relative tolerance. × 0% f [Hz] = 1.6 × 0%

Image capture



Quiz3 第2题 原始页面 3/4

完整题目、填写值、上传回答与系统记录；相邻页有重叠

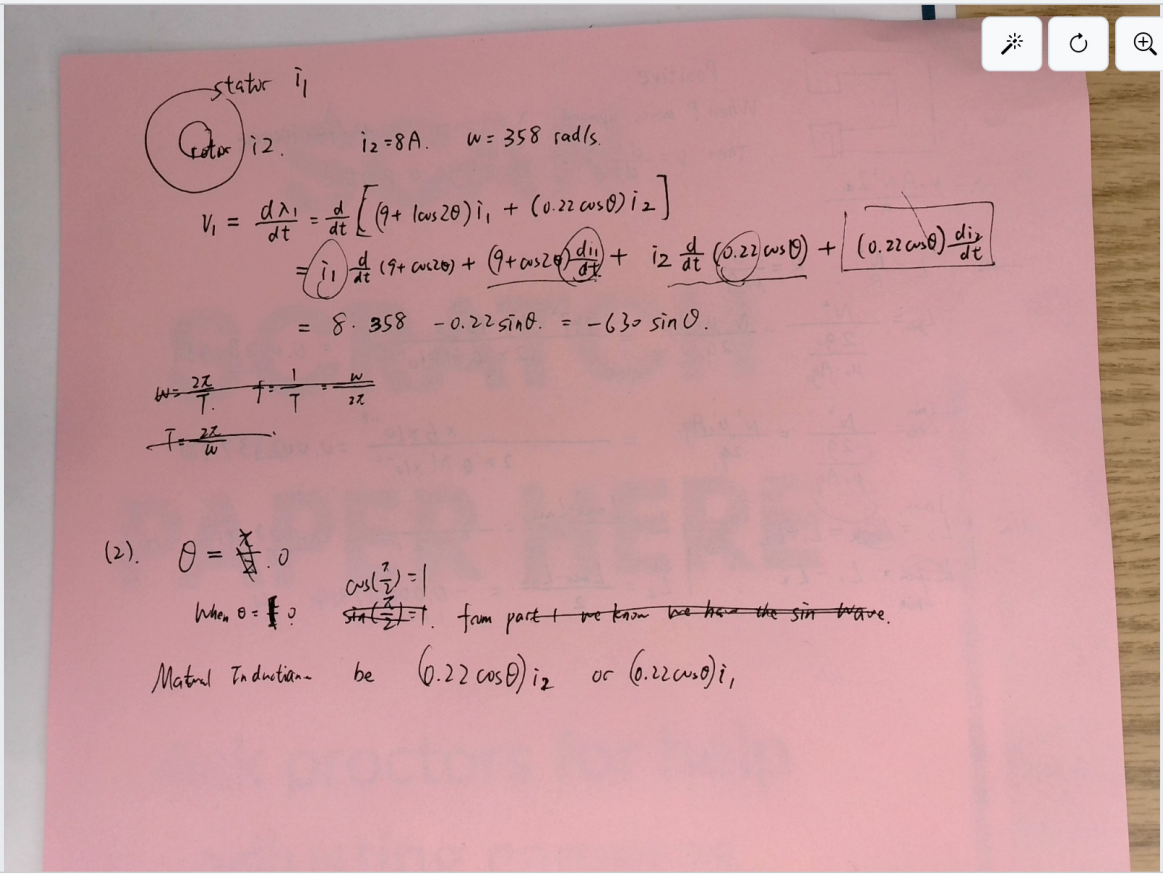
Submitted answer 2

yitaoma2@illinois.edu submitted at 2026-10-20 20:46:31 (CDT) **saved, not graded**

Hide

$|\overline{V}_1|$ [V] = 630 f [Hz] = 1.6

Image capture



Submitted answer 1

yitaoma2@illinois.edu submitted at 2026-10-20 20:42:19 (CDT) **saved, not graded**

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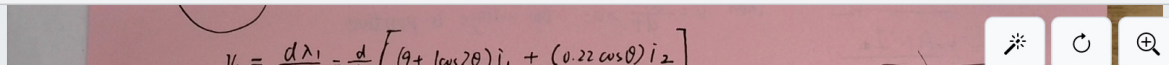
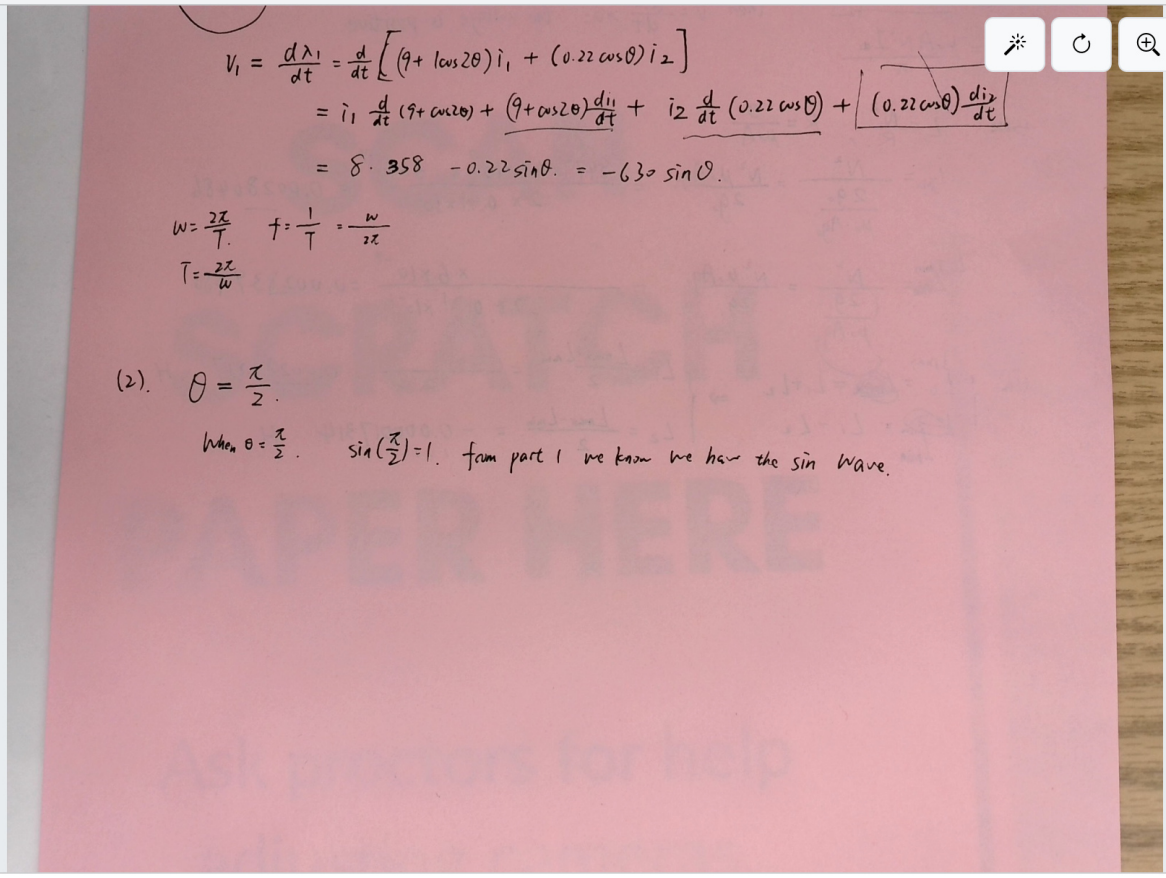
$|\overline{V}_1|$ [V] = 630 f [Hz] = 1.6

Image capture



Quiz3 第2题 原始页面 4/4

完整题目、填写值、上传回答与系统记录；相邻页有重叠



$$V_1 = \frac{d\lambda_1}{dt} = \frac{d}{dt}\left[(9+\cos 2\theta) i_1 + (0.22\cos\theta) i_2\right]$$

$$= i_1 \frac{d}{dt}(9+\cos 2\theta) + (9+\cos 2\theta)\frac{di_1}{dt} + i_2 \frac{d}{dt}(0.22\cos\theta) + (0.22\cos\theta)\frac{di_2}{dt}$$

$$= 8.358 \quad -0.22\sin\theta. \quad = -63\sin\theta.$$

$$w = \frac{2\pi}{T} \qquad f = \frac{1}{T} = \frac{w}{2\pi}$$

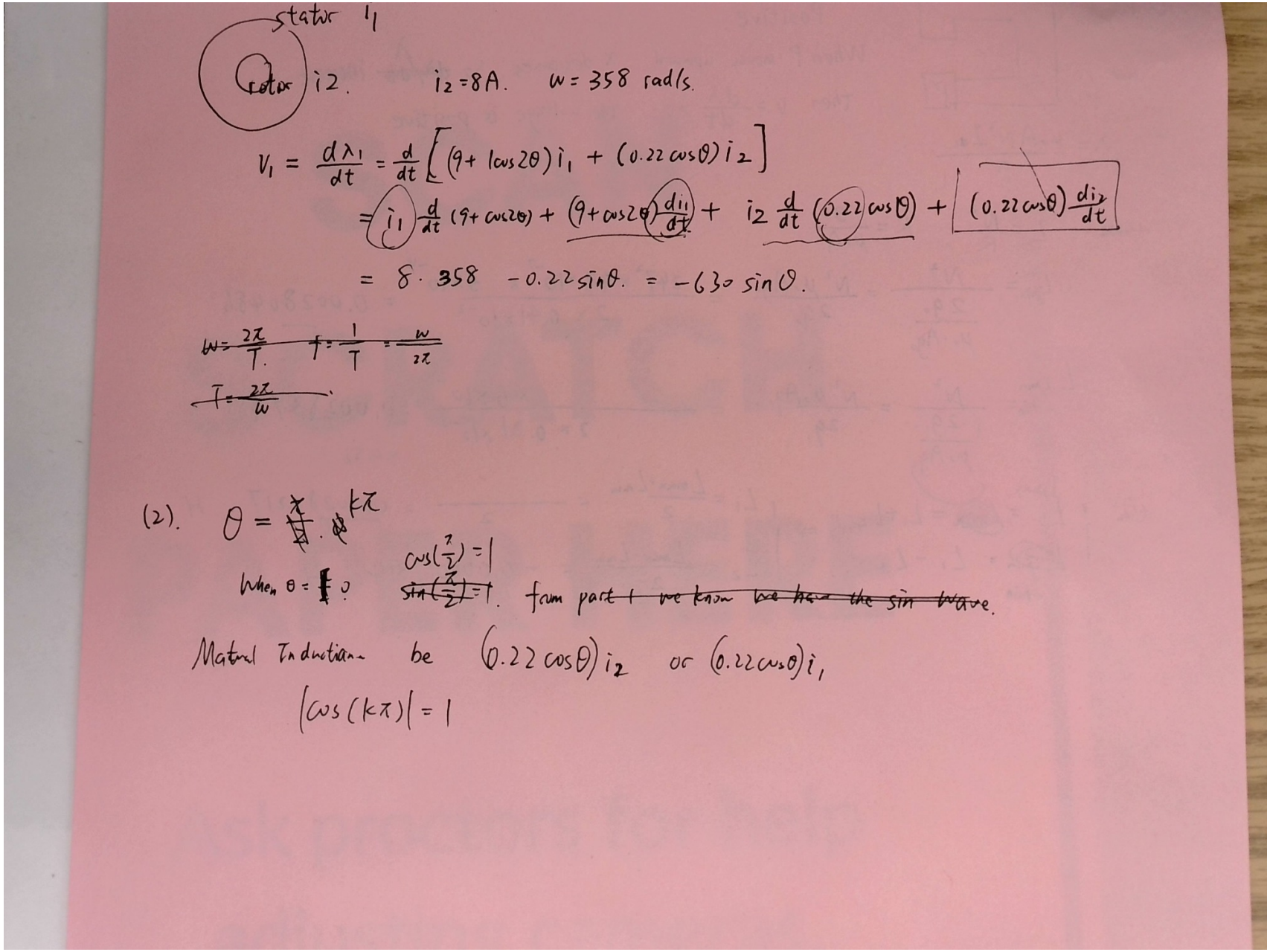
$$T = \frac{2\pi}{w}$$

(2). $\theta = \frac{\pi}{2}$.

When $\theta = \frac{\pi}{2}$. $\sin(\frac{\pi}{2}) = 1$. from part 1 we know we have the sin wave.

Quiz3 第2题 你的完整手写回答 1/3

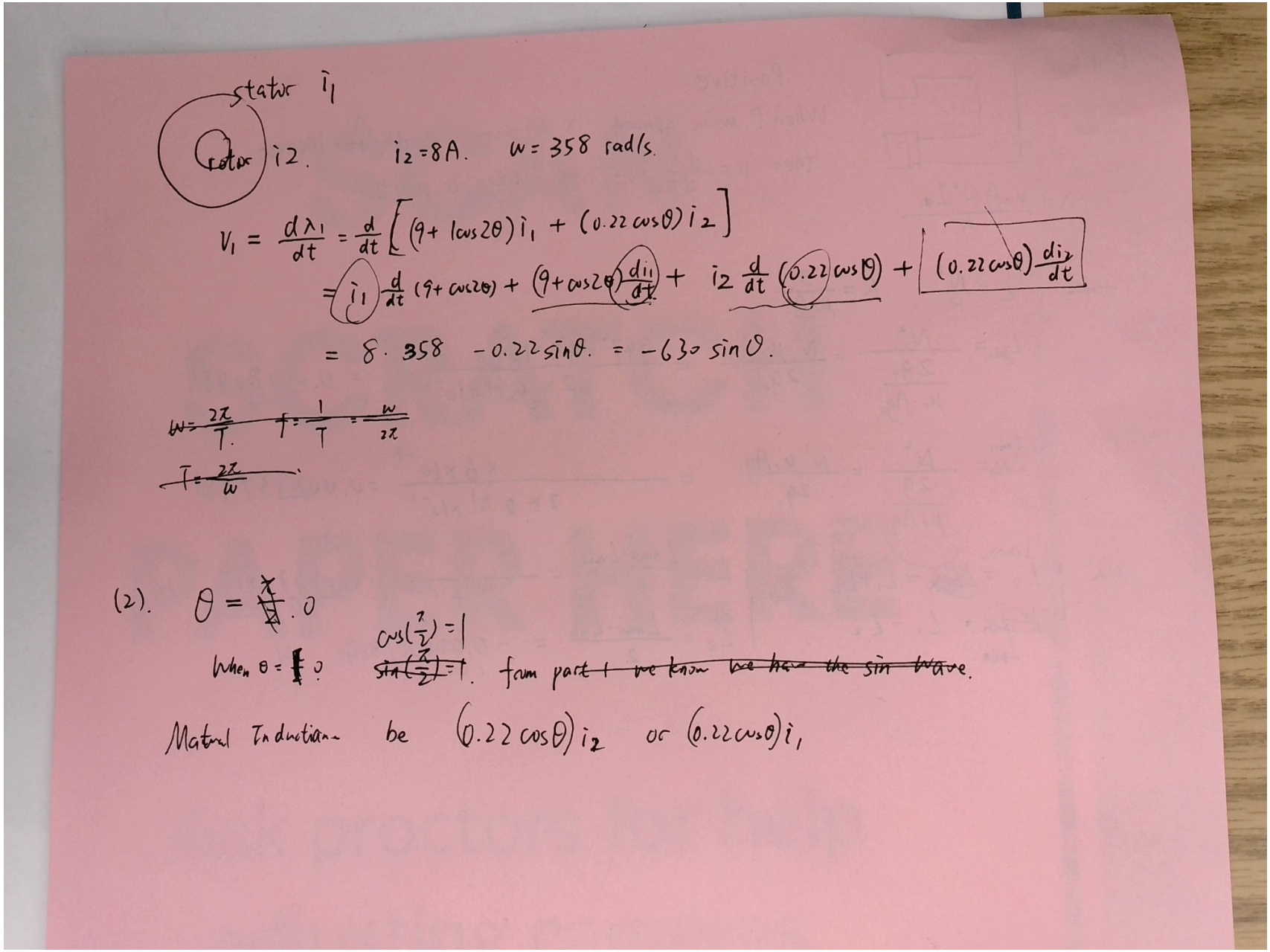
当前最终回答原图



Quiz3 Q2 | Full uploaded image | 2000 x 1500 px

Quiz3 第2题 你的完整手写回答 2/3

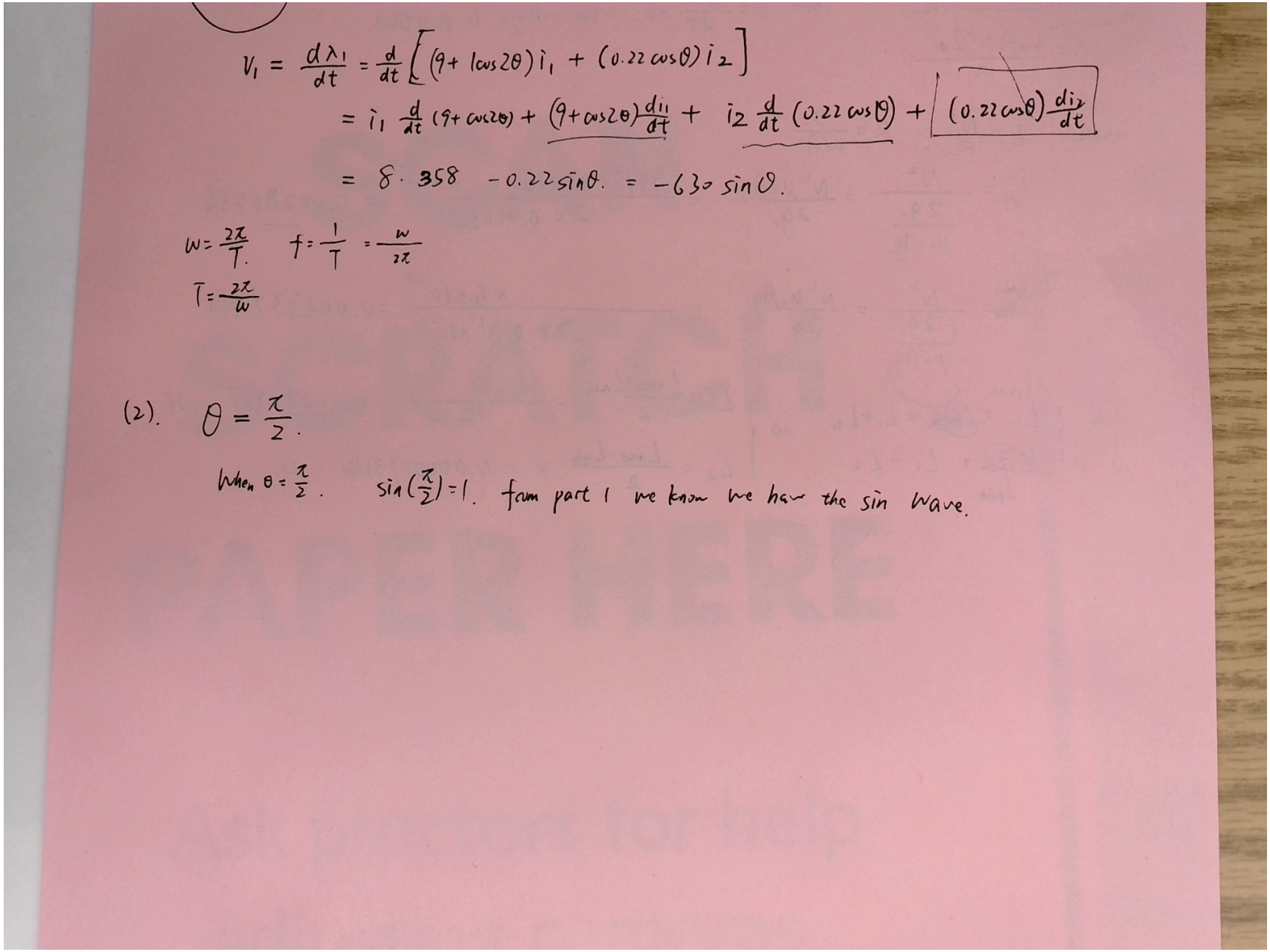
历史回答原图 版本2



Quiz3 Q2 | Full uploaded image | 2000 x 1500 px

Quiz3 第2题 你的完整手写回答 3/3

历史回答原图 版本3



$$V_1 = \frac{d\lambda_1}{dt} = \frac{d}{dt}\left[(9+\cos 2\theta) i_1 + (0.22\cos\theta) i_2\right]$$

$$= i_1 \frac{d}{dt}(9+\cos 2\theta) + (9+\cos 2\theta)\frac{di_1}{dt} + i_2 \frac{d}{dt}(0.22\cos\theta) + (0.22\cos\theta)\frac{di_2}{dt}$$

$$= 8.358 \quad -0.22\sin\theta. \quad = -63\sin\theta.$$

$\omega = \frac{2\pi}{T}$ $\quad f = \frac{1}{T} = \frac{\omega}{2\pi}$

$T = \frac{2\pi}{\omega}$

(2). $\theta = \frac{\pi}{2}$.

When $\theta = \frac{\pi}{2}$. $\quad \sin(\frac{\pi}{2}) = 1$. from part 1 we know we have the sin wave.

Quiz3 Q2 | Full uploaded image | 2000 x 1500 px

第2题 开路发电机与互感最大值

开路消项、恒流励磁、链式求导、RMS、频率、带符号互感的最大值

柴油机驱动转子，端口1为定子、端口2为转子，角度为 θ。转子接 8 A 恒流直流源，角速度 358 rad/s，定子开路。给定：

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} 9+\cos(2\theta) & 0.22\cos\theta \\ 0.22\cos\theta & 66+34\cos(2\theta) \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}.$$

第一问（5分）求开路定子正弦电压相量的幅值和频率，**明确要求 RMS**。第二问（3分）问互感在哪个角度最大，并解释；只为理由给分。

$$\begin{aligned} i_1 &\equiv 0, \quad i_2 = 8\ \text{A}, \quad \theta(t) = 358t + \theta_0, \\ \lambda_1 &= (9+\cos 2\theta)(0) + (0.22\cos\theta)(8) = 1.76\cos\theta, \\ v_1 &= \frac{d\lambda_1}{dt} = -630.08\sin(358t+\theta_0)\ \text{V}, \\ V_{1,\text{rms}} &= \frac{630.08}{\sqrt{2}} \approx 445.534\ \text{V}, \\ f &= \frac{358}{2\pi} \approx 56.9775\ \text{Hz}. \end{aligned}$$

系统显示 **446 V、57.0 Hz**；原输入为 **630 V、1.6 Hz**。630 V 对应峰值，不能直接当 RMS。早期回忆中的447 V、58 Hz不再作为原题事实。

互感 M(θ)＝0.22cosθ，最大正值为0.22 H，在 **θ＝2kπ（k为整数）**。θ＝(2k＋1)π 时为最小负值，虽然绝对值同样最大。自感的二倍角所在整项已被开路条件消去，不决定输出频率。

Quiz3 第3题 原始页面 1/2

完整题目、填写值、上传回答与系统记录；相邻页有重叠

Question 3

NOTE: Full credit (100%) for this question is 4 points: 2 points for the correct numerical answer(s) and 2 points for the correct conceptual answer. Points for the conceptual answer will be added manually.

(2 pts) Consider an electromechanical system with energy given as $W_m = x\lambda^2$

Plot the system on a $\lambda - i$ plane for x = 2 cm and x = 5 cm. Use the same plane to plot both graphs. **Submit your answer on scratch paper.**

(2 pts) The co-energy of a device is given as $W'_m = \frac{i^6}{4x}$

What is the flux linkage λ when $i = 0.8$ A and $x = 0.41$ cm?.

λ[Wb turns]=	119.883	?	✓ 100%

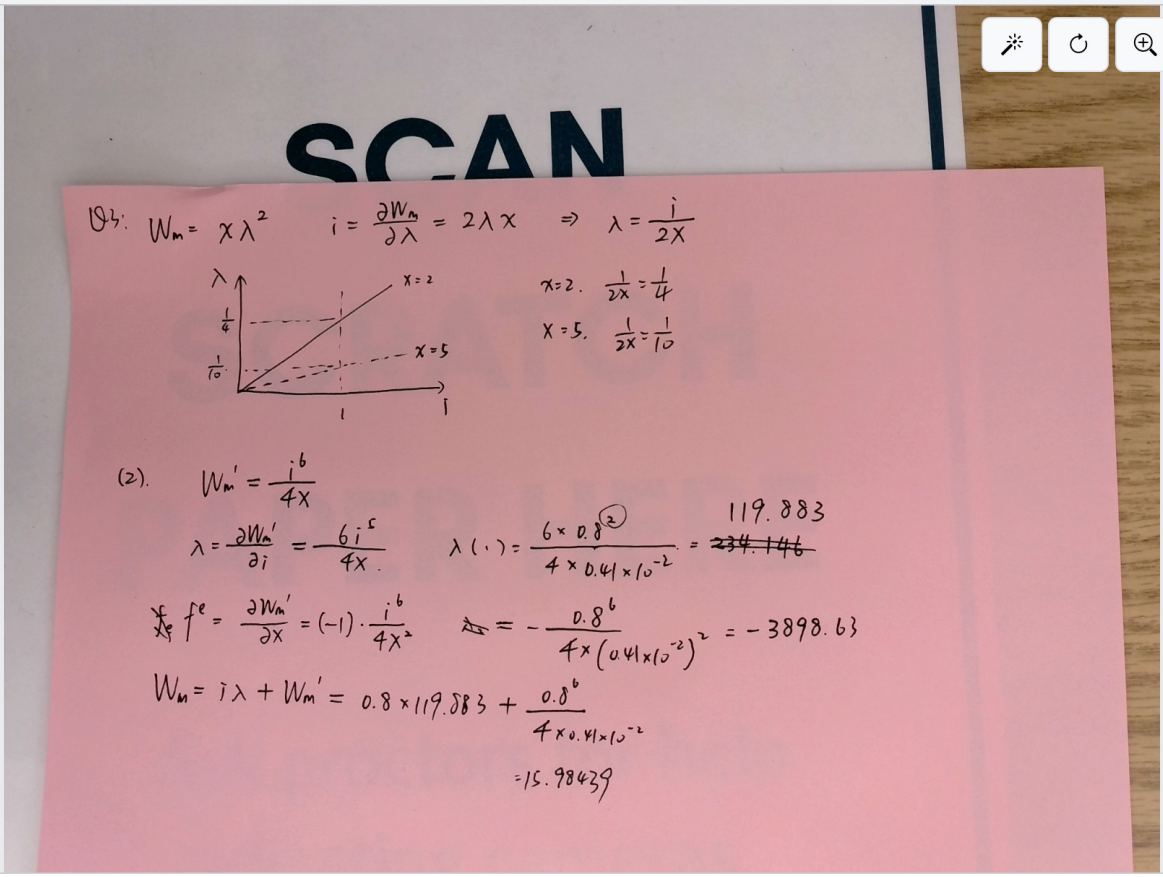
What is the force of electric origin $f^e(i, x)$ for these conditions?

f^e[N]=	-3898.63	?	✓ 100%

What is the energy W_m for the same conditions?

W_m[J]=	111.89079	?	✗ 0%

Image capture



Correct answer

Quiz3 第3题 原始页面 2/2

完整题目、填写值、上传回答与系统记录；相邻页有重叠

Correct answer

λ [Wb turns]= 119.882926829 f^e [N]= −3898.63176681 W_m [J]= 79.9219512195

Feedback from the Course Staff Hide

correct conceptual answer Awarded partial credit see gradescope(scaling is a little off but point justification reasoning is on gradescope, great work!)

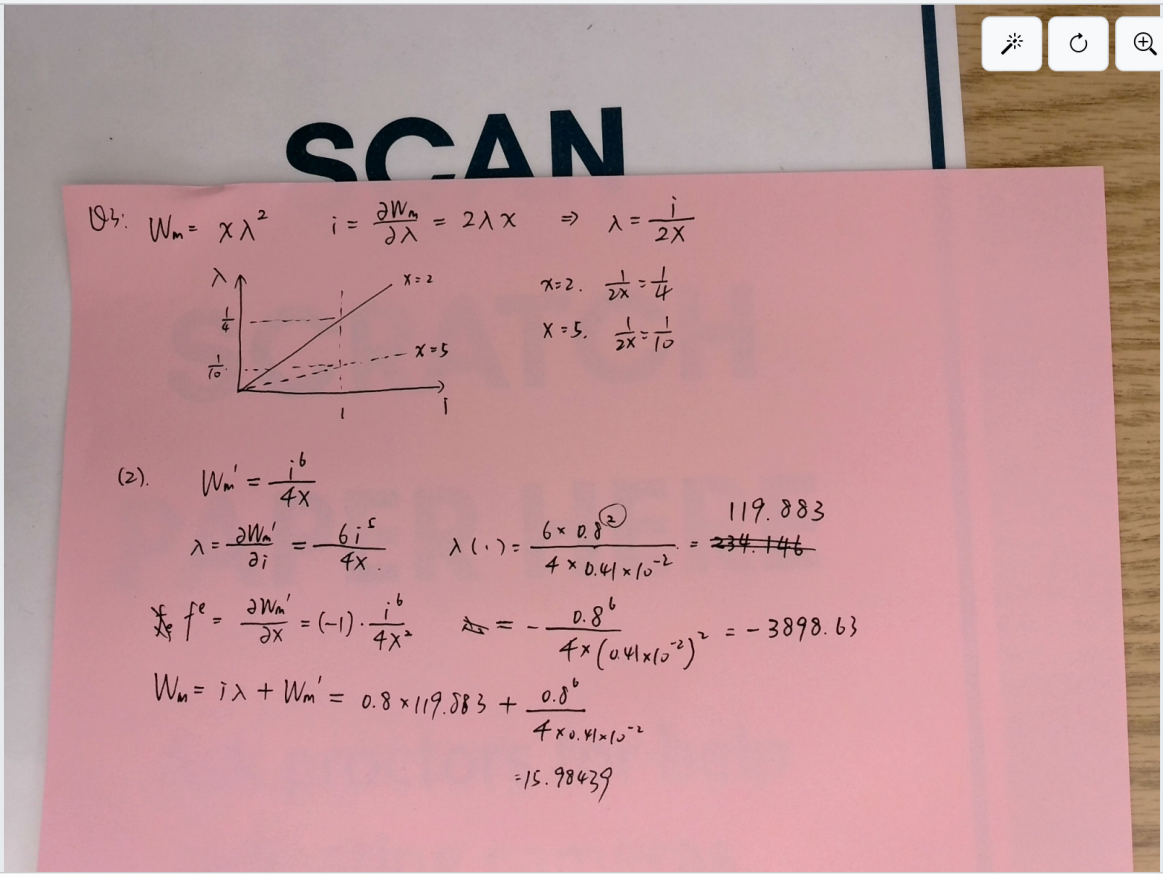
Submitted answer
yitaoma2@illinois.edu submitted at 2026-10-20 20:33:30 (CDT)
manual grading: 108%
auto-grading: 66%
Hide

λ [Wb turns]= 119.883 The correct answer used for grading was 119.88292682926834. ✓ 100%

f^e [N]= −3898.63 The correct answer used for grading was -3898.6317668054753. ✓ 100%

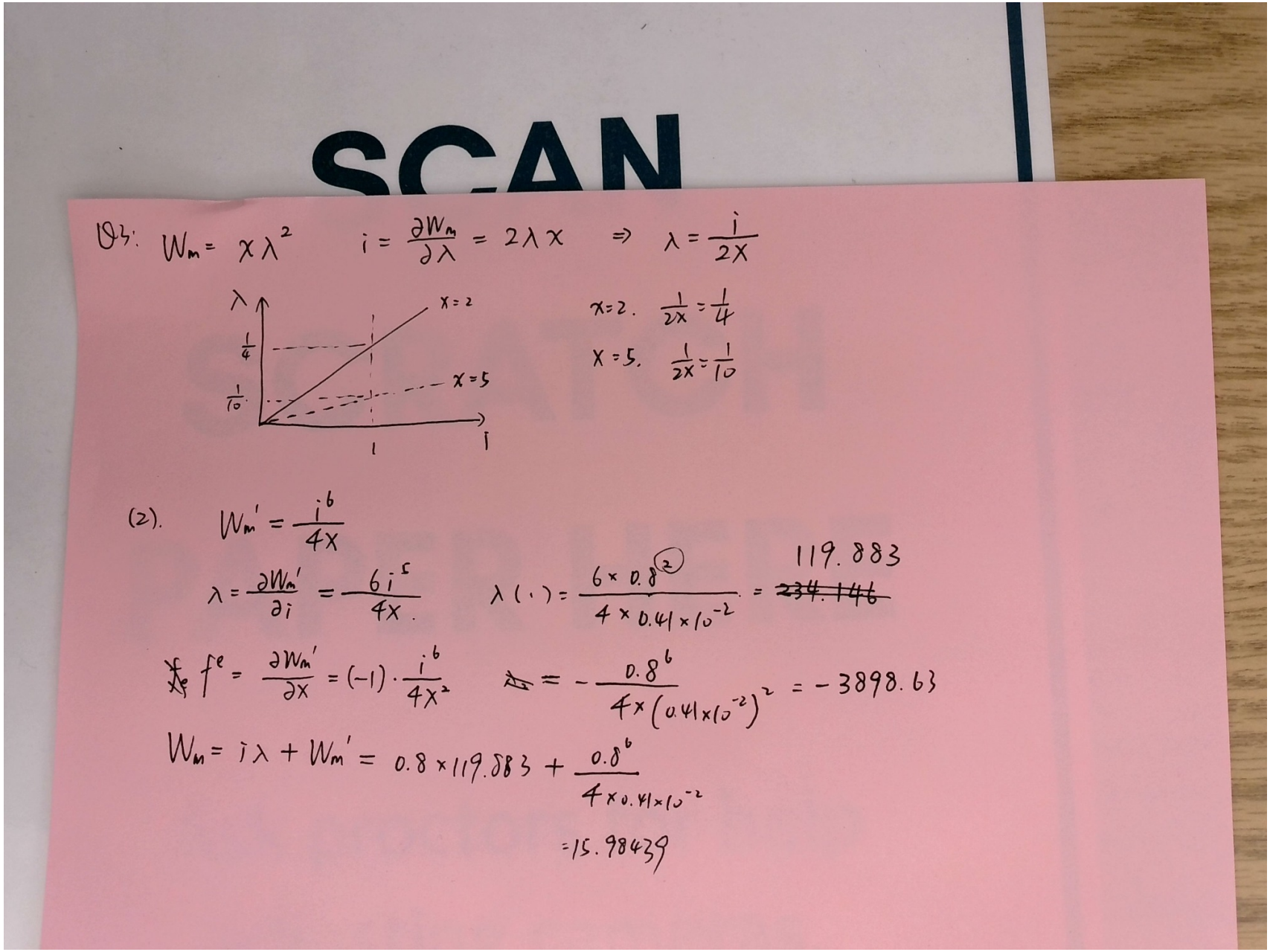
W_m [J]= 111.89079 ✕ 0%

Image capture



Quiz3 第3题 你的完整手写回答 1/1

当前最终回答原图



Quiz3 Q3 | Full uploaded image | 2000 x 1500 px

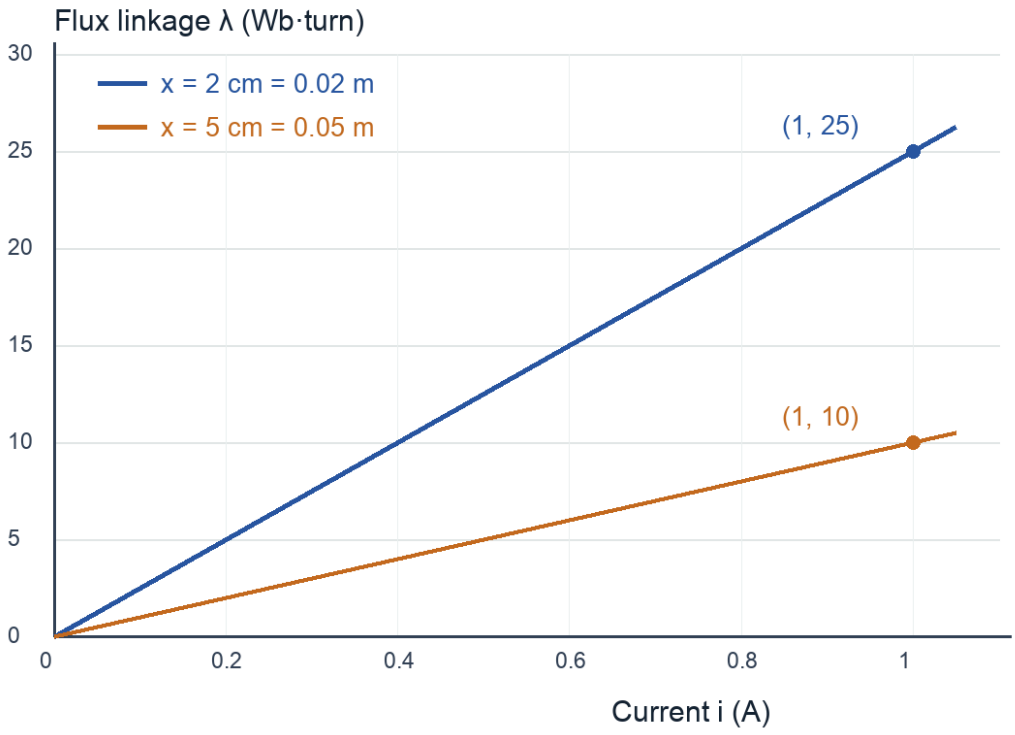
第3题 能量图线与非线性共能

能量偏导画 λ–i 图、共能偏导求磁链与力、能量共能相减

第一问（2分） 给能量 $W_m = x\lambda^2$，要求在同一 λ–i 平面画出 x = 2 cm、5 cm 两条图线。

$$i = \left.\frac{\partial W_m}{\partial \lambda}\right|_x = 2x\lambda \quad \Rightarrow \quad \lambda = \frac{i}{2x}.$$

横轴 i、纵轴 λ，两条直线均过原点。用米代入后，两条斜率分别为25 H、10 H，因此2 cm图线更陡。这问不要求代入第二问的非线性模型。



第二问（2分） 另给共能如下，i = 0.8 A、x = 0.41 cm，求磁链、力和能量。

$$W'_m(i,x) = \frac{i^6}{4x}, \qquad x = 0.0041 \text{ m}.$$

$$\begin{aligned}
\lambda &= \left.\frac{\partial W'_m}{\partial i}\right|_x = \frac{3i^5}{2x} = \frac{3(0.8)^5}{2(0.0041)} \approx 119.883 \text{ Wb}\cdot\text{turn},\\
f^e &= \left.\frac{\partial W'_m}{\partial x}\right|_i = -\frac{i^6}{4x^2} = -\frac{(0.8)^6}{4(0.0041)^2} \approx -3898.63 \text{ N},\\
W'_m &= \frac{(0.8)^6}{4(0.0041)} \approx 15.9844 \text{ J},\\
W_m &= i\lambda - W'_m = \frac{5i^6}{4x} = \frac{5(0.8)^6}{4(0.0041)} \approx 79.9220 \text{ J}.
\end{aligned}$$

负力表示沿 x 减小方向；能量公式的减号来自 $W_m + W'_m = i\lambda$。 该模型非线性，不能套两者相等的线性结论。系统显示磁链119.882926829、力−3898.63176681、能量79.9219512195。你的磁链与力已判对；原能量输入111.89079 J来自把共能相加。

ECE330 Quiz4

完整题目、你的完整回答与对应解答

共3题，逐题保留实际网页记录与上传手写原图。

每题按原页连续截图、完整手写图、解答的顺序排列。

长网页分段连续展示，相邻页留有重叠，防止截断文字。

手写图使用完整上传图片，未裁去纸面上的作答。

历史上传图片不同的版本均保留；相同图片不重复放大。

截图与数值记录来自你的已完成测验页面。

解答为独立解析，系统答案与反馈在原页截图中保留。

截图日期 2026年11月4日

Quiz4 第1题 原始页面 1/2

完整题目、填写值、上传回答与系统记录；相邻页有重叠

Question 1

NOTE: Full credit (100%) for this question is 7 points: all for the correct numerical answer(s).

(7 pts) The flux linkage in an energy-conservative electromechanical system is given by

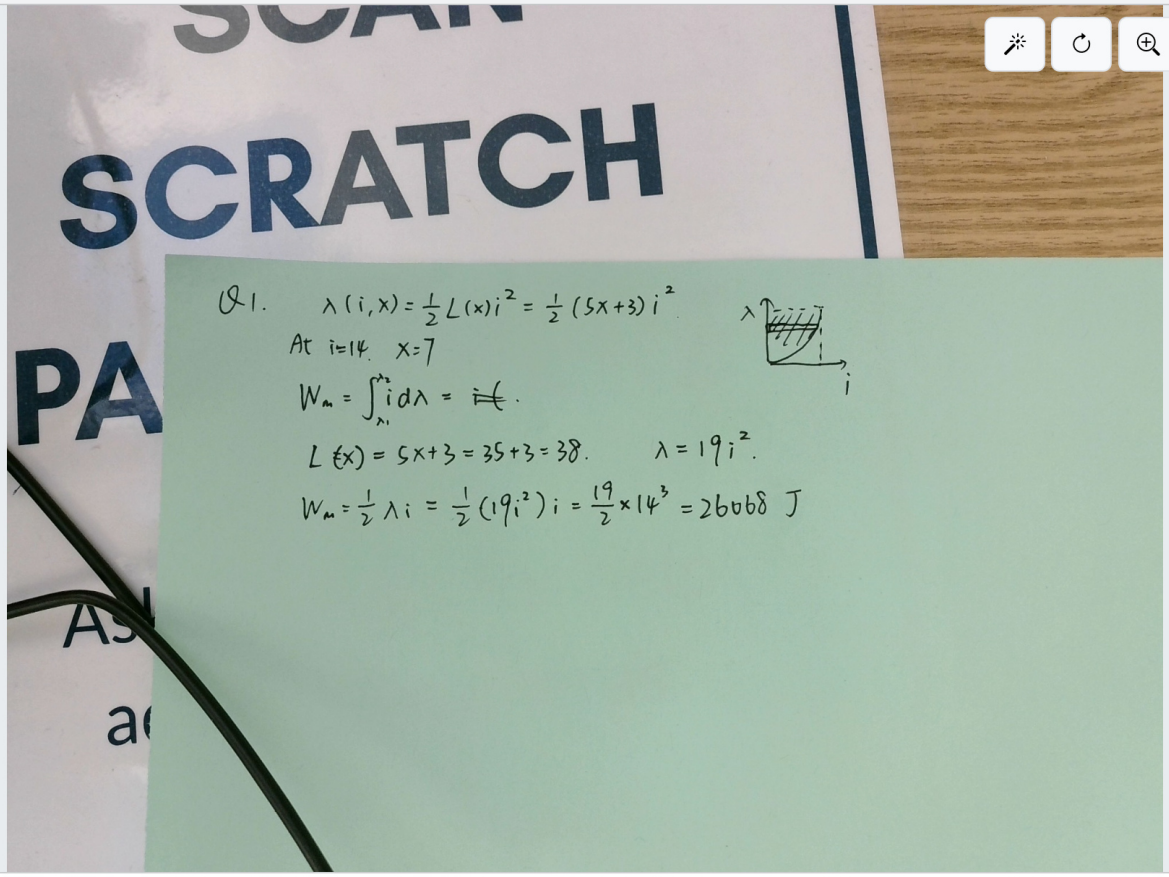
$$\lambda(i,x) = \frac{1}{2}L(x)i^2$$

where $L(x) = 5x + 3$ H/A. Here, x is the size of the gap in m, and i is the current applied to the coil in A.

Compute the energy (in Joules) in the system at $i = 14$ A and $x = 7$ m.

Wm [J]= 26068 ✕ 0%

Image capture



Correct answer

Wm [J]= 34757.3333333

Submitted answer

yitaoma2@illinois.edu submitted at 2026-11-03 19:06:38 (CST) 0%

Hide

Quiz4 第1题 原始页面 2/2

完整题目、填写值、上传回答与系统记录；相邻页有重叠

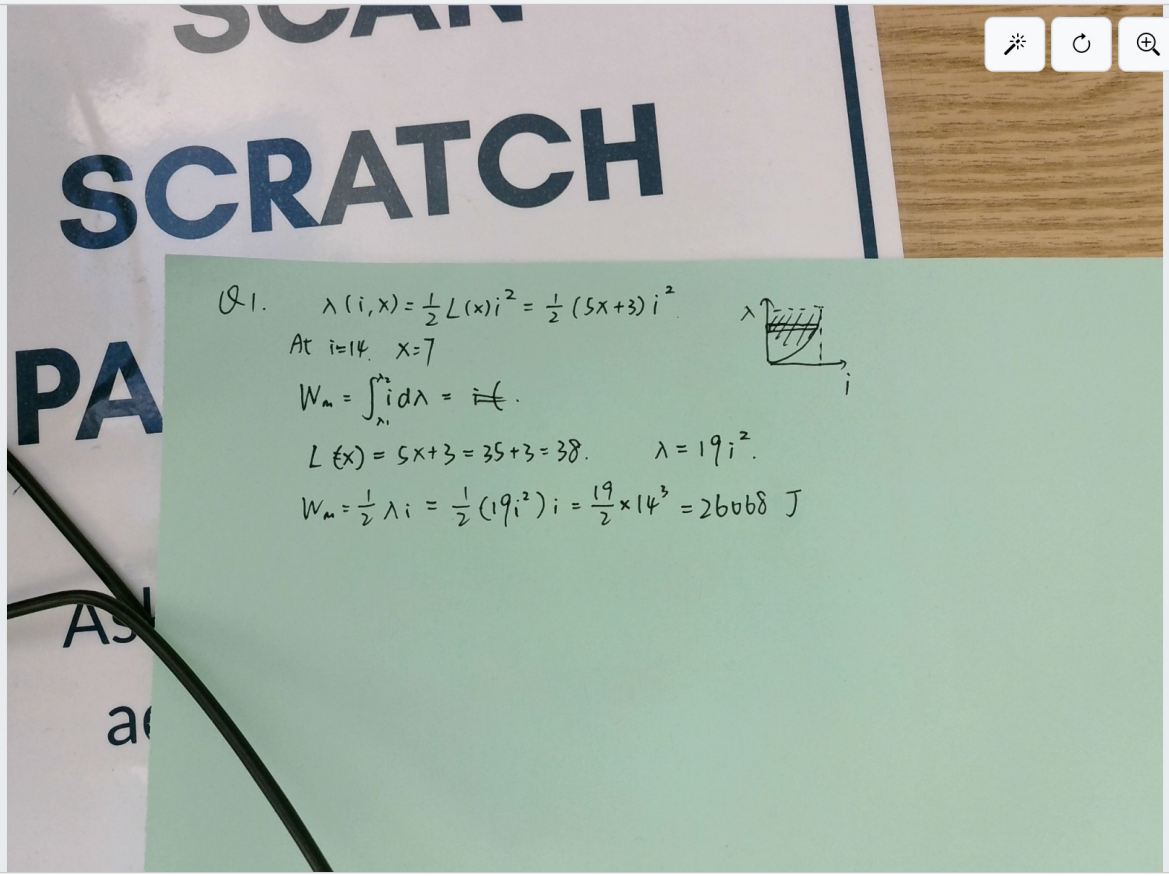
Submitted answer

yitaoma2@illinois.edu submitted at 2026-11-03 19:06:38 (CST) 0%

Hide

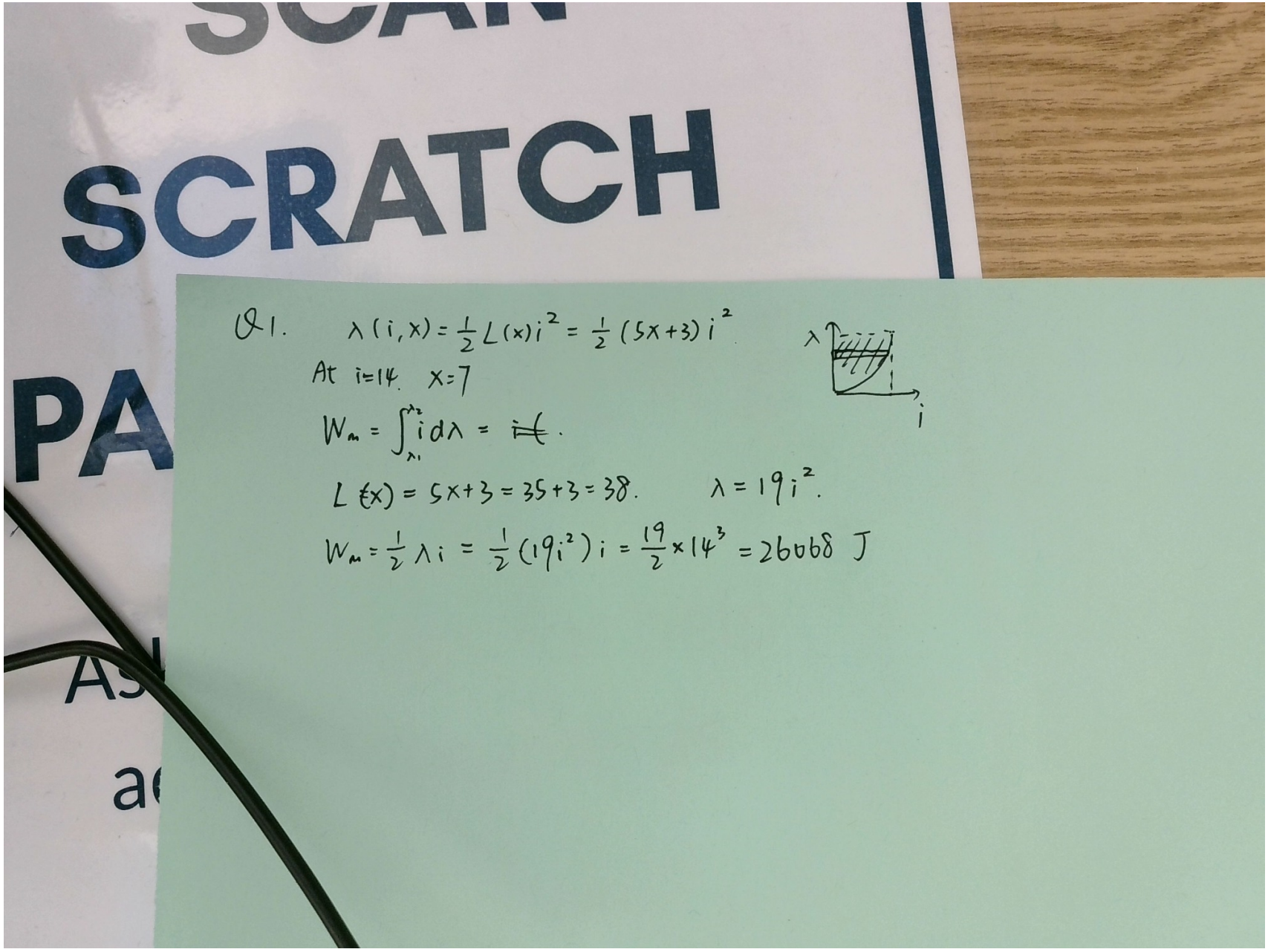
Wm [J]= 26068 | Your answer does not have precision within the specified relative tolerance. ✕ 0%

Image capture



Quiz4 第1题 你的完整手写回答 1/1

当前最终回答原图



Quiz4 Q1 | Full uploaded image | 2000 x 1500 px

第1题 非线性磁链的磁场能量

中文题意

保守机电系统满足 $\lambda(i,x)=\frac{1}{2}L(x)i^2$，其中 $L(x)=5x+3$，单位为 H/A；x 为以米计的气隙，i 为以安培计的线圈电流。求 $i=14\,\mathrm{A}$、$x=7\,\mathrm{m}$ 时的磁场能量 W_m，单位 J。

推导与结果

这里要求的是磁场能量，即在位置固定时对磁链积分 $W_m=\int i\,d\lambda$。题给磁链正比于电流的平方，不能套用线性模型的 $\frac{1}{2}i\lambda$。先固定 $x=7$，求出本构系数：

$$L(7)=5(7)+3=38\ \mathrm{H/A}.$$

将题给磁链对电流求导，得到积分中的磁链增量：

$$d\lambda=L(x)i\,di.$$

把这个表达式放回能量定义，积分变量用 $\hat{i}$ 区分于最终电流：

$$W_m=\int_0^{14}\hat{i}\,[38\hat{i}\,d\hat{i}]=38\left[\frac{\hat{i}^3}{3}\right]_0^{14}$$

$$=\frac{38(14)^3}{3}=34757.3333\ \mathrm{J}.$$

核对：$\lambda=\frac{1}{2}(38)(14)^2=3724\,\mathrm{Wb\cdot turn}$；共能 $W'_m=38(14)^3/6=17378.6667\,\mathrm{J}$。两者之和为 $i\lambda=52136\,\mathrm{J}$。你填入的 $26068\,\mathrm{J}$ 对应 $\frac{1}{2}i\lambda$，这一步把非线性本构当成了线性本构。系统显示的正确值为 $34757.3333333\,\mathrm{J}$。

最终英文作答版

At fixed $x=7\,\mathrm{m}$, $L=5(7)+3=38\,\mathrm{H/A}$ and $d\lambda=38i\,di$. Therefore,

$$W_m=\int_0^{14} i\,d\lambda=\int_0^{14}38i^2\,di=\frac{38(14)^3}{3}=34757.3333\ \mathrm{J}.$$

The expression $W_m=\frac{1}{2}i\lambda$ does not apply because λ is quadratic in i.

Quiz4 第2题 原始页面 1/2

完整题目、填写值、上传回答与系统记录；相邻页有重叠

Question 2

NOTE: Full credit (100%) for this question is 6 points: 3 points for the correct numerical answer(s) and 3 points for the correct conceptual answer. Points for the conceptual answer will be added manually.

The flux linkage for a given electromechanical system is given as $\lambda = \frac{Ai}{x - x_0}$ where $A = 0.059$ Hm and $x_0 = 0.01$ m

$$\begin{bmatrix} & a & b & c & d & a \\ i & 0 & i_b & i_c & 0 & 0 \\ \lambda & 0 & 5 & \lambda_c & 0 & 0 \\ x & 0.038 & 0.038 & 0.022 & 0.022 & 0.038 \end{bmatrix}$$

The system is operated in a closed cycle $a \to b \to c \to d \to a$. x is held constant along $a \to b$ and $c \to d$ and i is held constant over $b \to c$ and $d \to a$.

(3 pts) Plot the system on a $\lambda - i$ plane. **Submit your answer on scratch paper.**

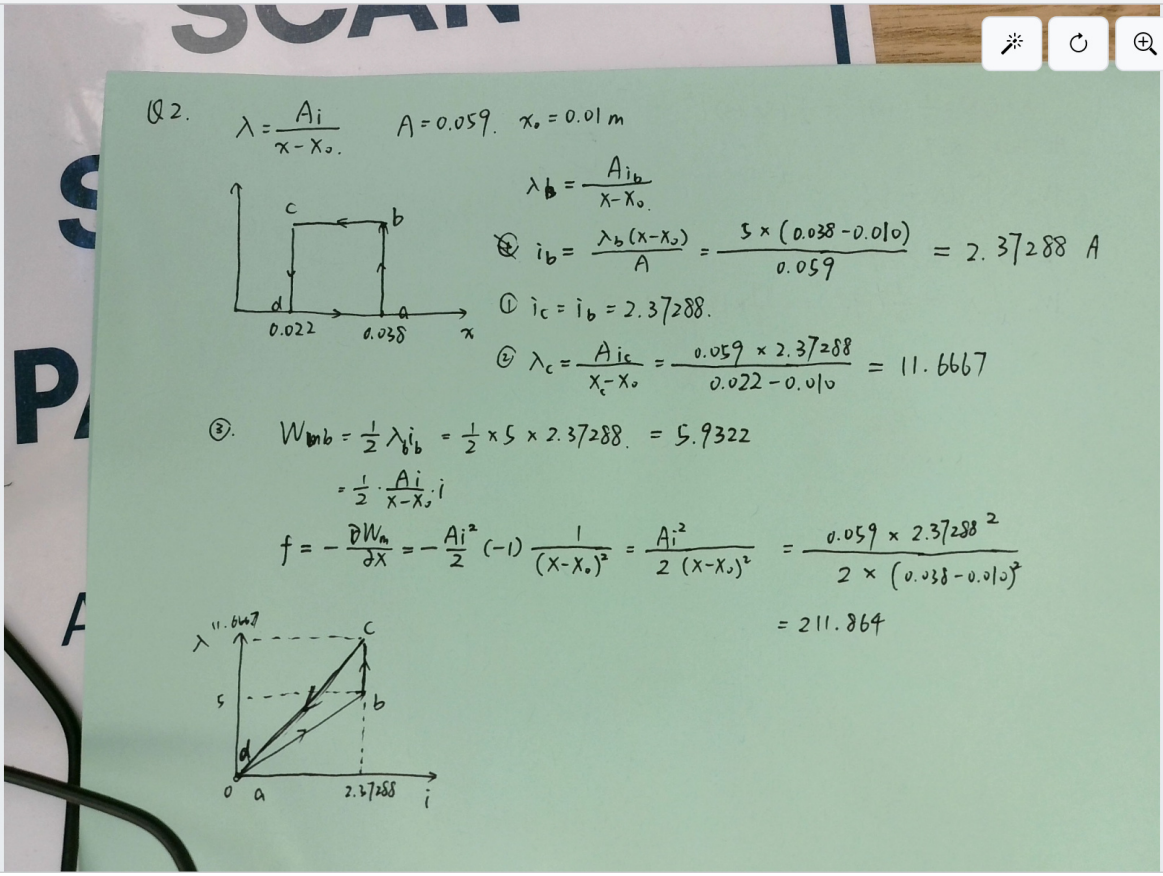
(3 pts) What is i_c and λ_c?

i_c[A]=	2.37288	?	✓ 100%
λ_c[Wb turns]=	11.6667	?	✓ 100%

What is energy stored in the coupling field and the force of electric origin at point b?

W_{m_b}[J]=	5.9322	?	✓ 100%
f_b^e[N]=	211.864	?	✗ 0%

Image capture



Quiz4 第2题 原始页面 2/2

完整题目、填写值、上传回答与系统记录；相邻页有重叠

Correct answer

i_c [A]= 2.37288135593 λ_c [Wb turns]= 11.6666666667 W_{m_b} [J]= 5.93220338983 f_b^e [N]= −211.86440678

Feedback from the Course Staff Hide

Conceptual: Mostly incorrect answer (-2 pts)

Submitted answer
yitaoma2@illinois.edu submitted at 2026-11-03 19:17:43 (CST)
manual grading: 33%
auto-grading: 75%
Hide

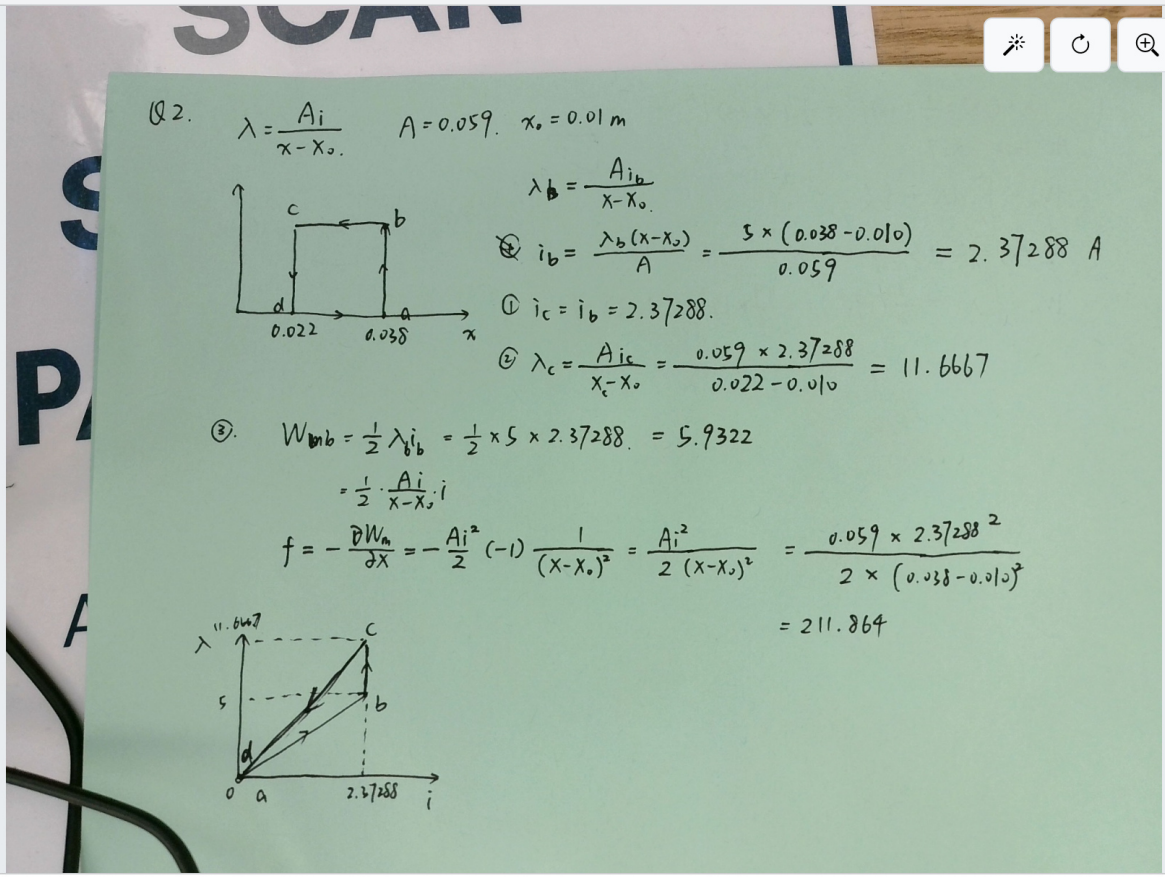
i_c [A]= 2.37288 | The correct answer used for grading was 2.3728813559322033. ✓ 100%

λ_c [Wb turns]= 11.6667 | The correct answer used for grading was 11.666666666666666. ✓ 100%

W_{m_b} [J]= 5.9322 | The correct answer used for grading was 5.932203389830508. ✓ 100%

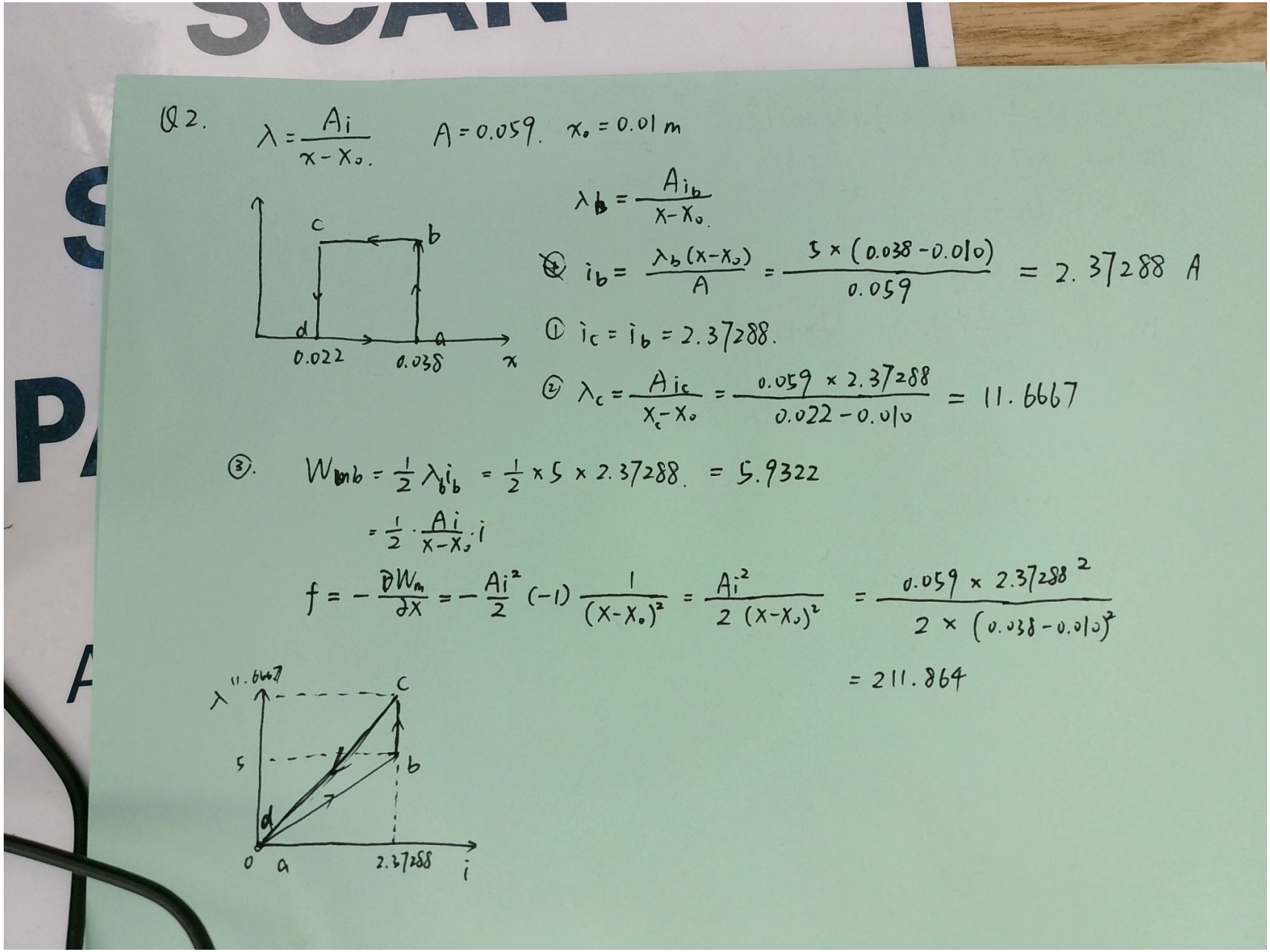
f_b^e [N]= 211.864 ✗ 0%

Image capture



Quiz4 第2题 你的完整手写回答 1/1

当前最终回答原图



Quiz4 Q2 | Full uploaded image | 2000 x 1500 px

第2题 闭合循环与固定磁链求力

中文题意

系统满足 $\lambda = Ai/(x - x_0)$，其中 $A = 0.059\,\mathrm{H\,m}$、$x_0 = 0.010\,\mathrm{m}$。按 $a \to b \to c \to d \to a$ 运行：$a \to b$ 和 $c \to d$ 固定位置，$b \to c$ 和 $d \to a$ 固定电流。状态表给出 $a: (i, \lambda, x) = (0, 0, 0.038)$，$b: (i_b, 5, 0.038)$，$c: (i_c, \lambda_c, 0.022)$，$d: (0, 0, 0.022)$，最后回到 a。在同一 λ–i 平面画循环，求 i_c、λ_c，以及 b 点的磁场能量和电磁力。

先由恒流段连接状态

先用 b 点的磁链和位置解电流，再用 $b \to c$ 恒流把这个电流带到 c 点：

$$i_b = \frac{\lambda_b(x_b - x_0)}{A} = \frac{5(0.038 - 0.010)}{0.059} = 2.37288135593\ \mathrm{A}.$$

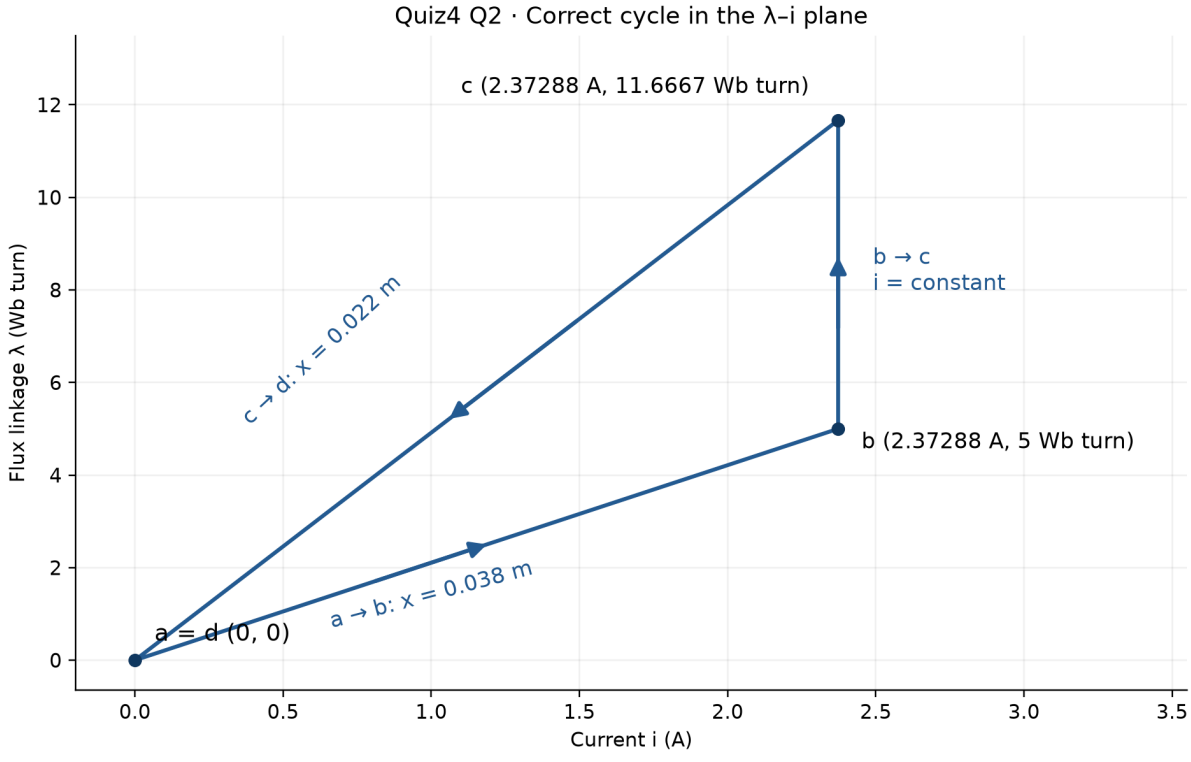
$$i_c = i_b = 2.37288135593\ \mathrm{A}.$$

将这个电流放回 c 点本构：

$$\lambda_c = \frac{0.059(2.37288135593)}{0.022 - 0.010} = 11.6666666667\ \mathrm{Wb \cdot turn}.$$

循环图与方向

横轴为电流 i，纵轴为磁链 λ。固定位置时是经过原点的直线；$b \to c$ 恒流，所以是竖直线。a 与 d 虽然位置不同，在这个平面都落在原点；$d \to a$ 不会产生另一条可见的线段。本问须手画或附上此图。



磁场能量与带符号电磁力

固定位置后，这道题的磁链对电流是线性的，所以能量可写为 $\frac{1}{2}i\lambda$：

$$W_{m,b}=\frac{1}{2}i_b\lambda_b=\frac{1}{2}(2.37288135593)(5)=5.93220338983\ \text{J}.$$

求力时把能量写成磁链和位置的函数。由 $i=\lambda(x-x_0)/A$，对磁链积分：

$$W_m(\lambda,x)=\int_0^{\lambda}\frac{\widehat{\lambda}(x-x_0)}{A}\,d\widehat{\lambda}=\frac{\lambda^2(x-x_0)}{2A}.$$

电磁力是固定磁链时能量对位置的负偏导。这里先固定 λ 求导，再代入 b 点：

$$f_b^e=-\left.\frac{\partial W_m}{\partial x}\right|_{\lambda}=-\frac{\lambda_b^2}{2A}=-\frac{5^2}{2(0.059)}=-211.864406780\ \text{N}.$$

负号表示力沿 x 减小的方向。你填写的电流、磁链和能量均与系统答案一致；力的大小一致，但输入少了负号。系统概念题反馈为“Mostly incorrect answer (−2 pts)”；此处保留该反馈原文，不推定更细的评分原因。

最终英文作答版

Since $b\to c$ has constant current,

$$i_c=i_b=\frac{5(0.038-0.010)}{0.059}=2.37288135593\ \text{A},$$

$$\lambda_c=\frac{0.059(2.37288135593)}{0.022-0.010}=11.6666666667\ \text{Wb}\cdot\text{turn}.$$

In the λ–i plane, draw $a=(0,0)$, $b=(2.37288136,5)$, $c=(2.37288136,11.6666667)$, and $d=(0,0)$, with arrows $a\to b\to c\to d$. The path $d\to a$ remains at the origin. Copy or attach the labeled cycle diagram above.

$$W_{m,b}=\frac{1}{2}(5)(2.37288135593)=5.93220338983\ \text{J}.$$

$$W_m(\lambda,x)=\frac{\lambda^2(x-x_0)}{2A},\qquad f_b^e=-\left.\frac{\partial W_m}{\partial x}\right|_{\lambda}=-\frac{5^2}{2(0.059)}=-211.864406780\ \text{N}.$$

The force acts in the negative x direction.

Quiz4 第3题 原始页面 1/2

完整题目、填写值、上传回答与系统记录；相邻页有重叠

Question 3

NOTE: Full credit (100%) for this question is 7 points: all for the correct numerical answer(s).

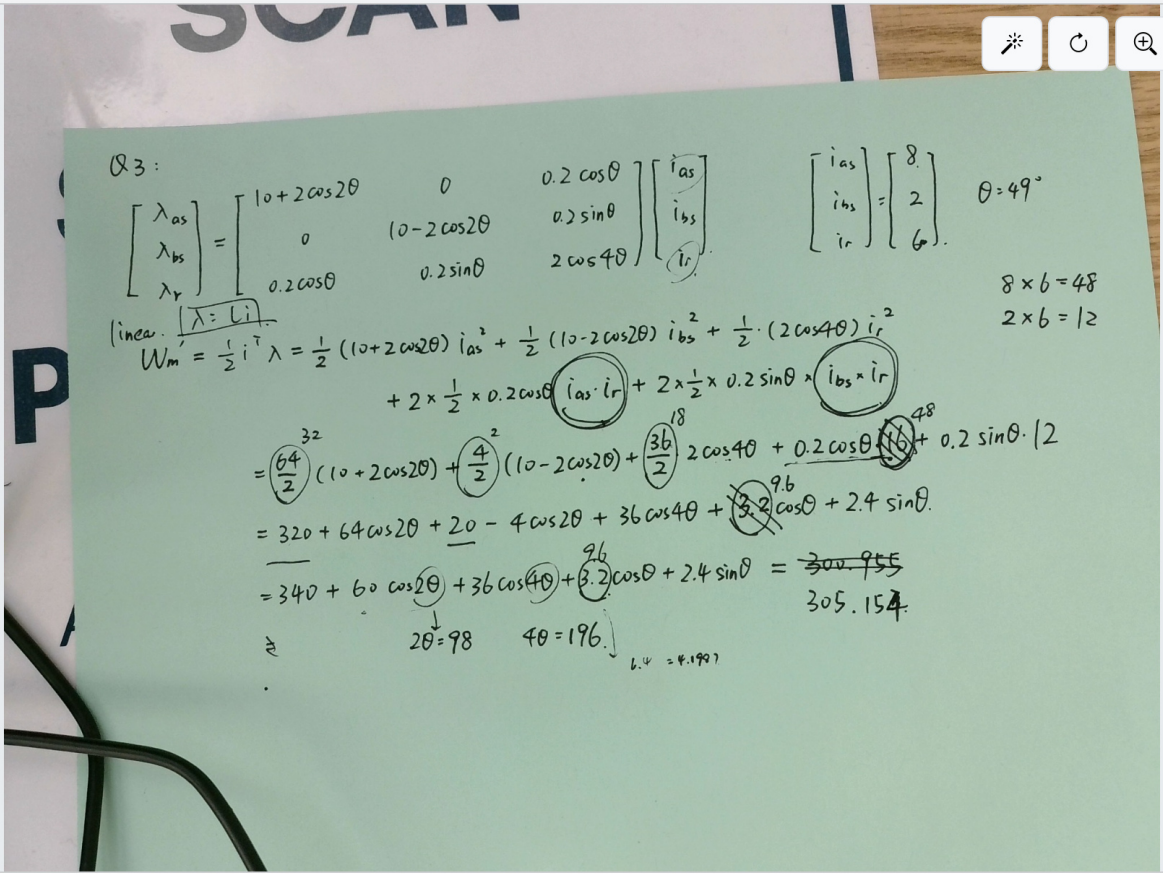
(7 pts) An electromechanical system has three electrical and one mechanical ports. The $\lambda - i$ relations are:

$$\begin{bmatrix}\lambda_{as}\\ \lambda_{bs}\\ \lambda_r\end{bmatrix} = \begin{bmatrix}10+2\cos(2\theta) & 0 & 0.2\cos(\theta)\\ 0 & 10-2\cos(2\theta) & 0.2\sin(\theta)\\ 0.2\cos(\theta) & 0.2\sin(\theta) & 2\cos(4\theta)\end{bmatrix}\begin{bmatrix}i_{as}\\ i_{bs}\\ i_r\end{bmatrix}$$

What is the co-energy at $i_{as} = 8$ A, $i_{bs} = 2$ A, $i_r = 6$ A and $\theta = 49°$?

W'_m[J]= 305.154 ✓ 100%

Image capture



Correct answer

W'_m [J]= 305.153662559

Submitted answer 3

yitaoma2@illinois.edu submitted at 2026-11-03 19:31:23 (CST) 100%

Hide

Quiz4 第3题 原始页面 2/2

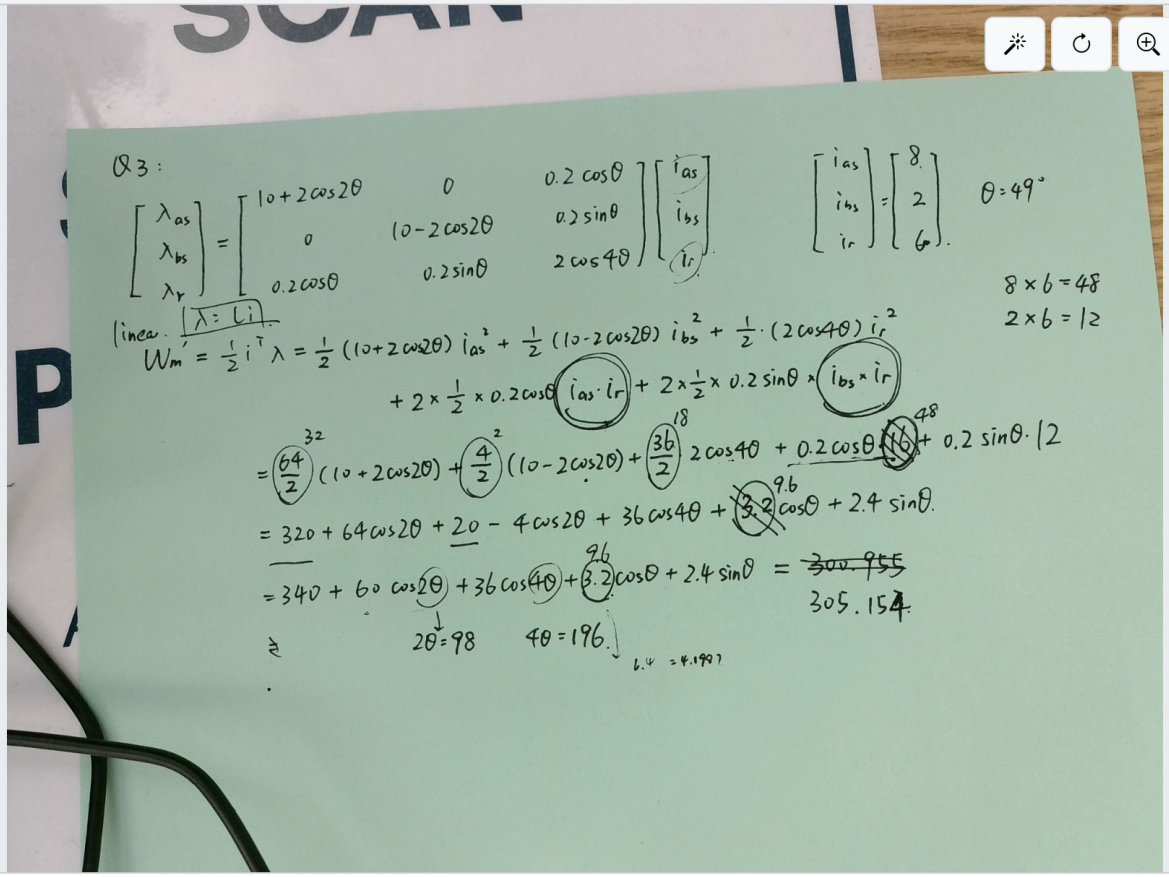
完整题目、填写值、上传回答与系统记录；相邻页有重叠

yitaoma2@illinois.edu submitted at 2026-11-03 19:31:23 (CST) 100%

Hide

W_m' [J]= 305.154 | The correct answer used for grading was 305.1536625594601. ✓ 100%

Image capture



Submitted answer 2

yitaoma2@illinois.edu submitted at 2026-11-03 19:26:05 (CST) saved, not graded

Show

Submitted answer 1

yitaoma2@illinois.edu submitted at 2026-11-03 19:25:56 (CST) invalid, not gradable

Hide

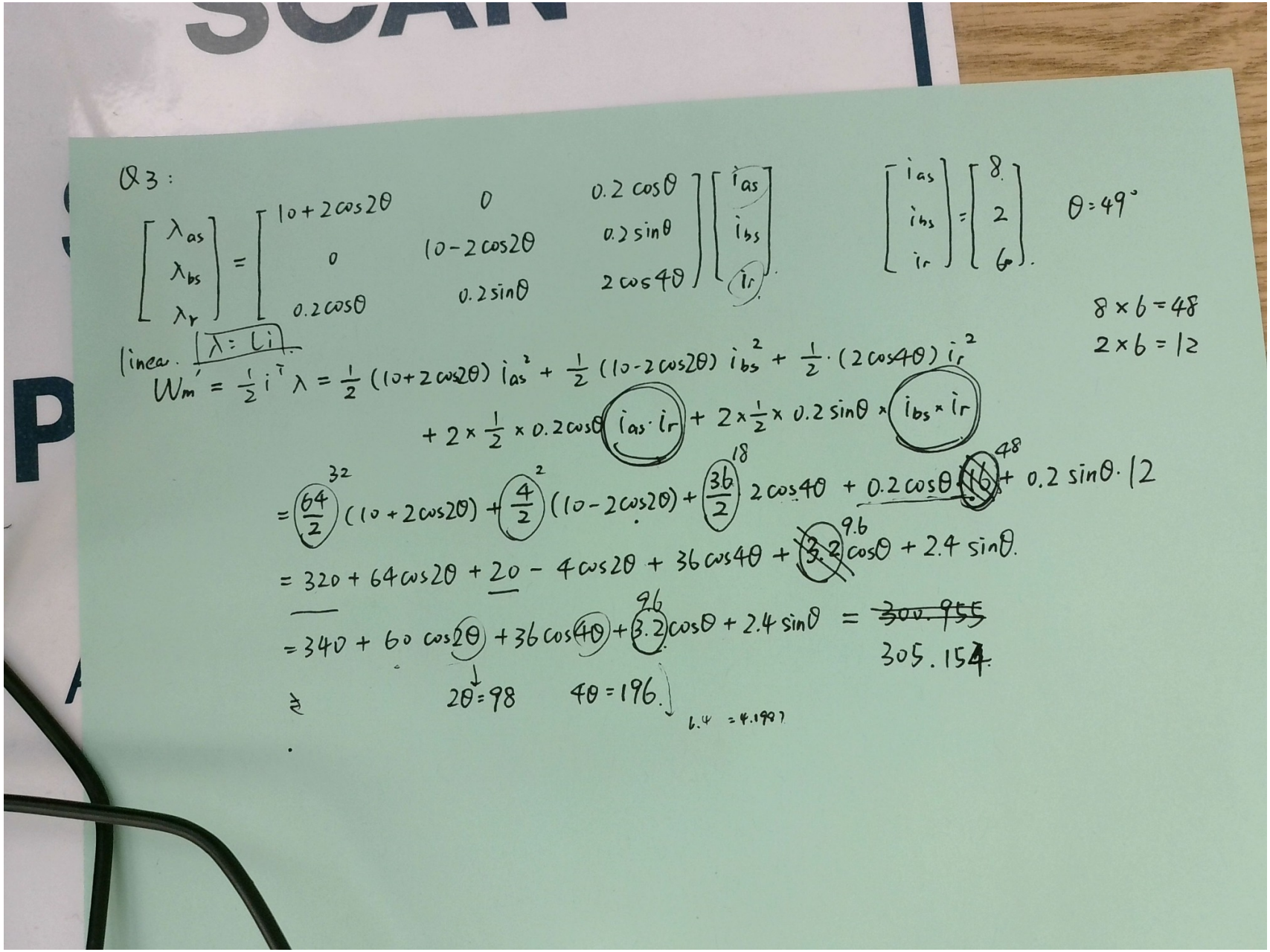
W_m' [J]= 300.955

Image capture

No image captured yet.
Use a clean sheet of paper.

Quiz4 第3题 你的完整手写回答 1/2

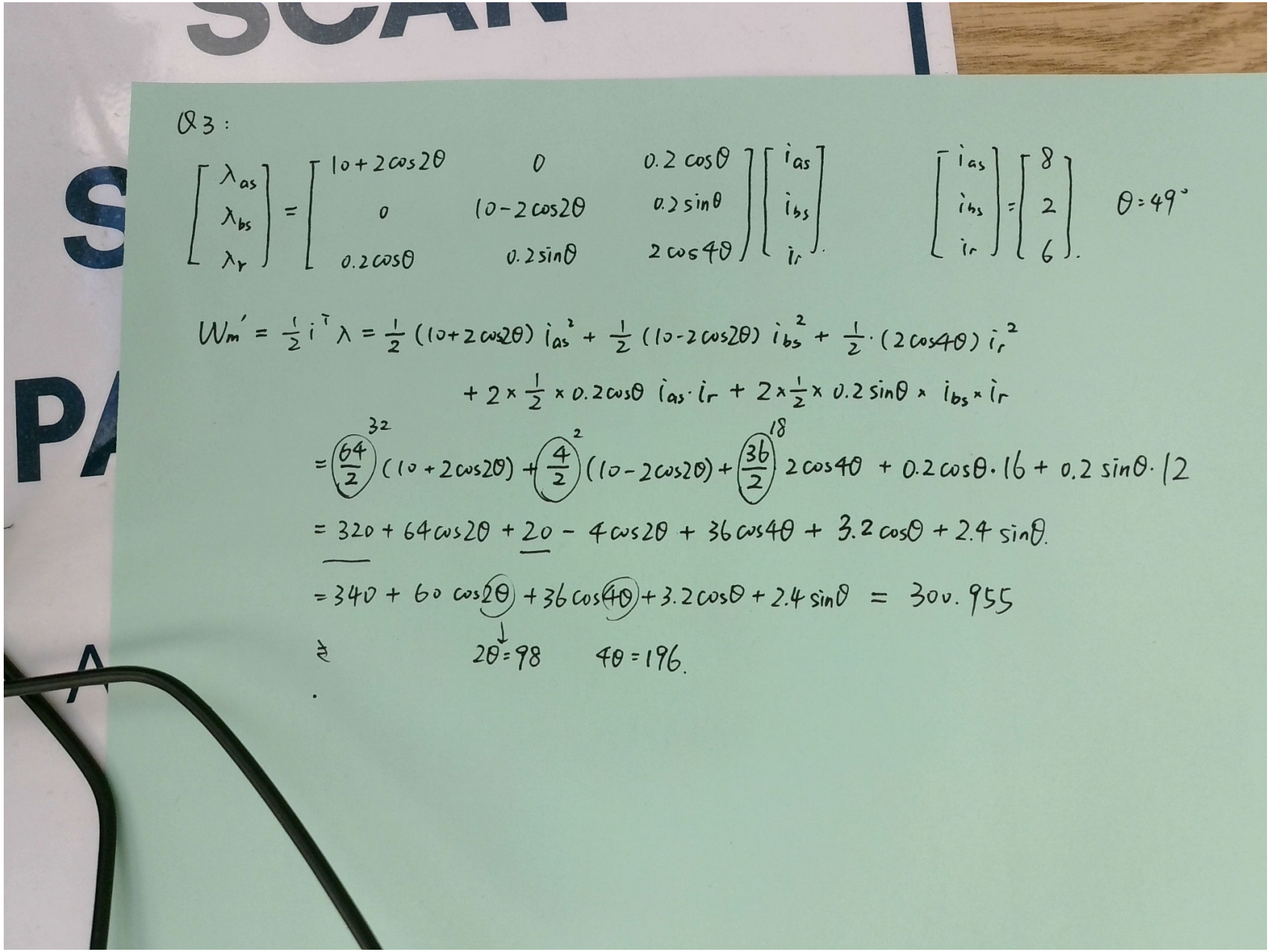
当前最终回答原图



Quiz4 Q3 | Full uploaded image | 2000 x 1500 px

Quiz4 第3题 你的完整手写回答 2/2

历史回答原图 版本2



Quiz4 Q3 | Full uploaded image | 2000 x 1500 px

第3题 三电端口的瞬时共能

中文题意

系统有三个电端口和一个机械端口，题给对称电感矩阵的对角项为 $10+2\cos 2\theta$、$10-2\cos 2\theta$、$2\cos 4\theta$；非零互感为 $0.2\cos\theta$ 和 $0.2\sin\theta$。求 $i_{as}=8\,\mathrm{A}$、$i_{bs}=2\,\mathrm{A}$、$i_r=6\,\mathrm{A}$、$\theta=49^\circ$ 时的共能 W'_m，单位 J。

从矩阵展开到数值

固定角度时，磁链对电流是线性的，且矩阵对称，因此 $W'_m=\frac{1}{2}\mathbf{i}^{\mathsf{T}}L\mathbf{i}$。对角项各带二分之一；每组非零互感在矩阵乘法里出现两次，与前面的二分之一相消。

$$
\begin{aligned}
W'_m &= \tfrac{1}{2}(10+2\cos 2\theta)i_{as}^2+\tfrac{1}{2}(10-2\cos 2\theta)i_{bs}^2 \\
&\quad +0.2i_{as}i_r\cos\theta+0.2i_{bs}i_r\sin\theta+i_r^2\cos 4\theta.
\end{aligned}
$$

把三电流和角度放回各自位置，使用 $2\theta=98^\circ$、$4\theta=196^\circ$：

$$
\begin{aligned}
W'_m &= \tfrac{1}{2}(10+2\cos 98^\circ)(8)^2+\tfrac{1}{2}(10-2\cos 98^\circ)(2)^2 \\
&\quad +0.2(8)(6)\cos 49^\circ+0.2(2)(6)\sin 49^\circ+(6)^2\cos 196^\circ \\
&= 340+60\cos 98^\circ+9.6\cos 49^\circ+2.4\sin 49^\circ+36\cos 196^\circ \\
&\approx 305.153662559\ \mathrm{J}.
\end{aligned}
$$

你的最终输入为 $305.154\,\mathrm{J}$，与系统答案一致。题目只要求此时的共能，不需要再求转矩。

最终英文作答版

At fixed θ, the flux–linkage model is linear in the currents and the inductance matrix is symmetric. Thus,

$$
\begin{aligned}
W'_m &= \tfrac{1}{2}\mathbf{i}^{\mathsf{T}}L\mathbf{i} \\
&= \tfrac{1}{2}(10+2\cos 98^\circ)(8)^2+\tfrac{1}{2}(10-2\cos 98^\circ)(2)^2 \\
&\quad +0.2(8)(6)\cos 49^\circ+0.2(2)(6)\sin 49^\circ+(6)^2\cos 196^\circ \\
&= 305.153662559\ \mathrm{J}.
\end{aligned}
$$